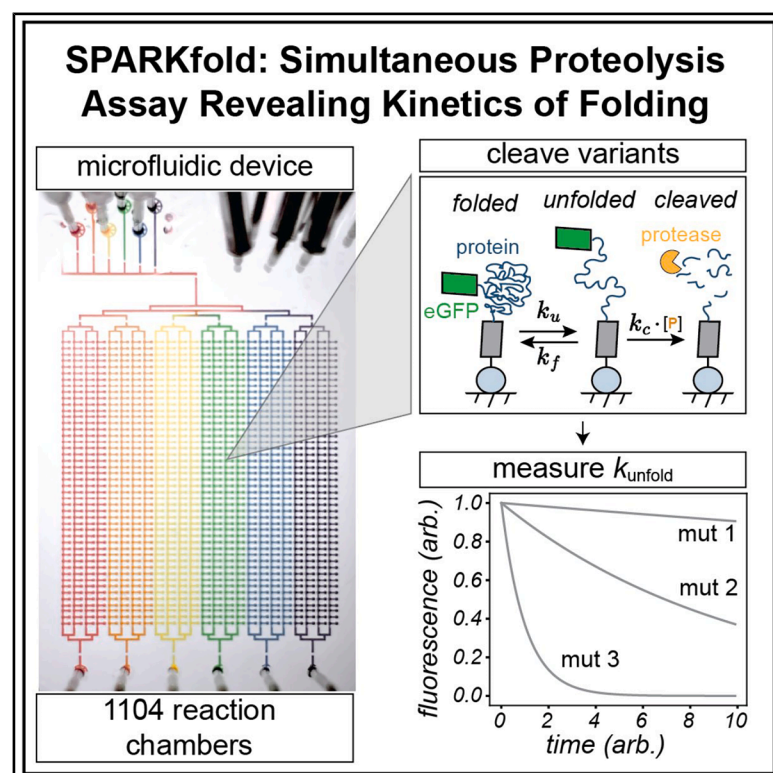


Quantifying protein unfolding kinetics with a high-throughput microfluidic platform

Graphical abstract



Authors

Beatriz Atsavaprane, Fanny Sunden, Daniel Herschlag, Polly M. Fordyce

Correspondence

pfordyce@stanford.edu

In brief

Atsavaprane et al. present a microfluidic platform that accurately quantifies unfolding kinetics for many proteins in parallel. After expressing and immobilizing EGFP-tagged variants within device chambers, introducing the thermolysin protease drives protein cleavage as they unfold. Monitoring the loss of EGFP over time via microscopy quantifies unfolding rates while using minimal reagents.

Highlights

- A microfluidic proteolysis assay enables accurate measurement of protein unfolding rates
- Devices have >1,000 chambers that can be programmed with different protein variants
- EGFP-tagged proteins are immobilized on device surfaces prior to on-chip proteolysis
- Quantifying and fitting remaining EGFP fluorescence over time can yield unfolding rates



Article

Quantifying protein unfolding kinetics with a high-throughput microfluidic platform

Beatriz Atsavaprane, ^{1,6} Fanny Sunden, ^{2,6} Daniel Herschlag, ² and Polly M. Fordyce ^{1,3,4,5,7,*}¹Department of Bioengineering, Stanford University, Stanford, CA 94305, USA²Department of Biochemistry, Stanford University, Stanford, CA 94305, USA³Department of Genetics, Stanford University, Stanford, CA 94305, USA⁴Sarafan ChEM-H Institute, Stanford University, Stanford, CA 94305, USA⁵Chan Zuckerberg Biohub, Stanford University, Stanford, CA 94305, USA⁶These authors contributed equally⁷Lead contact*Correspondence: pfordyce@stanford.edu<https://doi.org/10.1016/j.cels.2026.101681>

SUMMARY

Even after folding, proteins sample unfolded intermediates at risk of irreversible alteration (e.g., via proteolysis, aggregation, or posttranslational modification). Thus, kinetic stability impacts protein lifetime and abundance. However, we have very few measurements of unfolding rates, largely due to technical challenges. To address this, we developed SPARKfold (simultaneous proteolysis assay revealing kinetics of folding), a microfluidic platform to express, purify, and measure unfolding rate constants at high throughput via native proteolysis. We applied SPARKfold to determine unfolding rate constants for 1,104 protein samples comprising 31 dihydrofolate reductase orthologs with up to 78 chamber replicates each, providing statistical power to resolve subtle effects. SPARKfold rate constants for 5 constructs agreed with traditional measurements across a 150-fold range and provided information about the folding transition state via ϕ analysis. In future work, SPARKfold can reveal mutations that drive misfolding and aggregation and enable the rational design of kinetically hyperstable variants for industrial use. A record of this paper's transparent peer review process is included in the supplemental information.

INTRODUCTION

Most proteins must fold to function, so thermodynamic stability—favoring the folded state—is required. But even when this criterion is met, proteins can still repeatedly unfold and sample higher-free-energy conformations before refolding again,^{1,2} and at each excursion to an unfolded state, proteins risk a variety of potential irreversible alterations, including cleavage by proteases, misfolding into an inactive conformation, and aggregation.^{3–8} Some proteins prone to aggregation (e.g., transthyretin⁹) have evolved to be kinetically hyperstable with very slow unfolding rates, reducing the number of times they risk degradation or aggregation over cellular timescales.¹⁰ Many secreted proteins, in addition to requiring stability to function in harsh environments, have similarly evolved kinetic stability to avoid irreversible unfolding.^{10,11} For proteins with modest kinetic stabilities, time-dependent degradation can serve as a molecular timer (e.g., serpin protease inhibitors and heat shock transcription factors).^{12–17} In these cases, posttranslational modifications often alter unfolding rates; e.g., ubiquitination can increase unfolding rates to drive proteasomal degradation of target proteins,^{18,19} whereas phosphorylation and methylation can modulate kinetic stability by altering protein interactions and conformational flexibility.^{20,21}

In an energy landscape diagram, thermodynamic stability is visualized as the difference in free energy between the folded and unfolded states, and kinetic stability is visualized as the height of the transition-state barrier between them (Figure 1A); under conditions in which refolding is slow compared with rates of cleavage or aggregation, changes in kinetic stability alone can drive significant changes in protein abundance over time¹¹ (Figure 1B).

A variety of high-throughput methods have been developed to probe protein thermodynamic stability,^{22–26} but measuring unfolding rates for many proteins in parallel has remained a challenge. Current kinetic approaches (e.g., stopped-flow kinetics^{27–32} and single-molecule fluorescence resonance energy transfer[FRET] assays^{33–36}) typically explore folding pathways by monitoring changes in spectroscopic signals (e.g., fluorescence and circular dichroism) during chemical or thermal denaturation. However, these approaches are often limited by the time and cost associated with protein expression and purification. Labeling assays (e.g., hydrogen-deuterium exchange^{37–40}) can directly monitor folding and unfolding rates but require expensive instrumentation and are not easily scaled.

Quantitative native proteolysis, recently developed by the Marqusee and Park labs, presents a potentially scalable



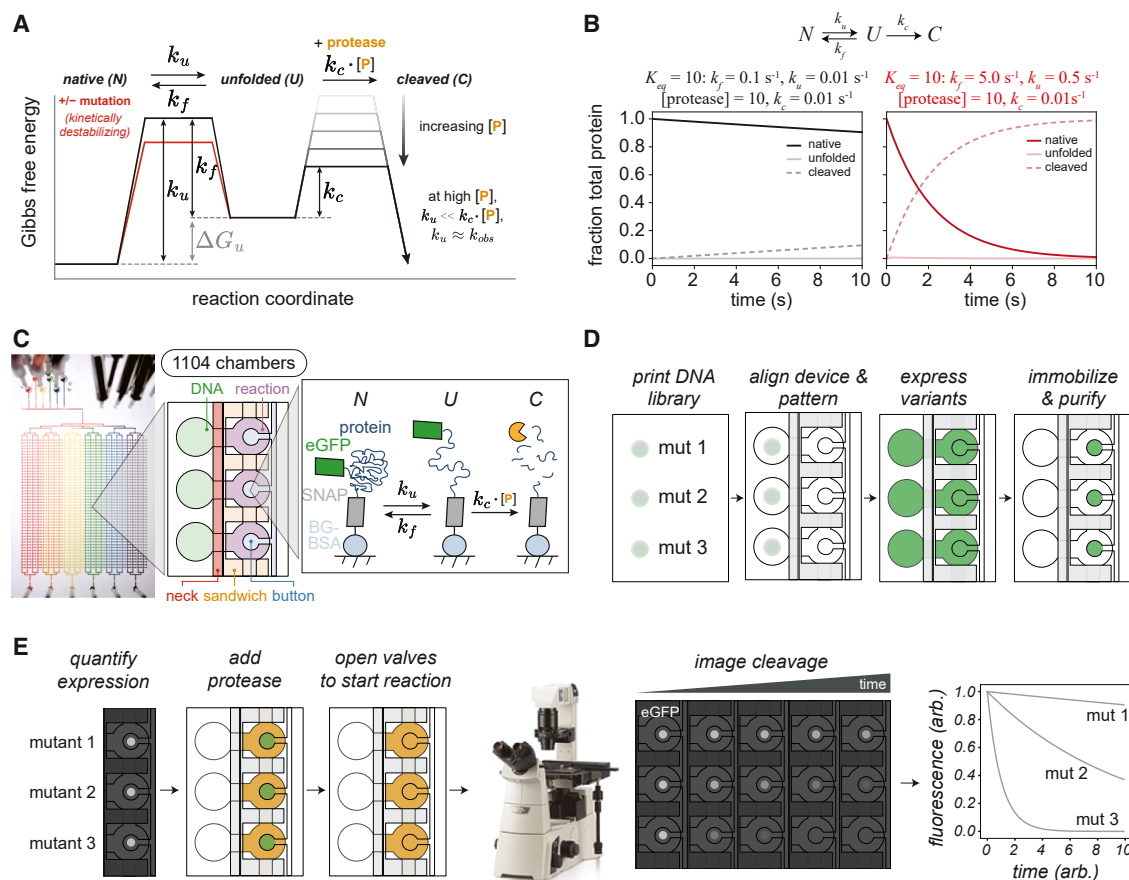


Figure 1. Protein folding and SPARKfold assay overview

(A) Gibbs free energy vs. reaction coordinate diagram for native, unfolded, and cleaved protein populations. Free energy of unfolding (ΔG_u) is the difference in Gibbs free energy between the native and unfolded states (gray). Height of the energy barrier between the native and unfolded states depends on the unfolding rate (k_u) and folding rate (k_f), which is altered by kinetically destabilizing mutations (red); height of the energy barrier between the unfolded and cleaved states is defined by the cleavage rate constant (k_c) and protease concentration ([P]). At high [P], k_u can be determined from the observed rate of proteolysis (k_{obs}), as carried out herein. At low [P], K_{eq} can be determined from k_{obs} (not carried out herein).

(B) Simulated data for the fractions of native (dark solid lines), unfolded (light solid lines), and cleaved (dashed lines) populations over time for two proteins with equal thermodynamic stability ($K_{eq} = 10$) but with either high ($k_f = 0.1 \text{ s}^{-1}$, $k_u = 0.01 \text{ s}^{-1}$; black) or low kinetic stability ($k_f = 5.0 \text{ s}^{-1}$, $k_u = 0.5 \text{ s}^{-1}$; red).

(C) Image and zoomed-in schematic of the SPARKfold device containing 1,104 chambers separated into 6 blocks (left). Each chamber has two compartments (DNA [green] and reaction [purple]) and 3 valves ("neck" [red], "sandwich" [orange], and "button" [blue]) (middle). Monitoring loss of fluorescence intensity over time for fluorescently tagged and surface-immobilized proteins exposed to protease provides a high-throughput method for measuring unfolding rates.

(D) Workflow for on-chip expression, immobilization, purification, and expression quantification.

(E) Simulated fluorescence images showing EGFP intensities after expression (left) and as a function of time after exposure to thermolysin protease (middle), yielding fluorescence-intensity decay curves (right).

alternative.^{41–43} As proteases cleave unfolded states, both thermodynamic and kinetic stability can be sensitively quantified by monitoring proteolytic cleavage over time under different conditions. For proteins that exhibit two-state folding between unfolded and native states (with folding rate k_f and unfolding rate k_u), K_{eq} can be obtained from the rate of proteolysis (k_{obs}), as the protein unfolds to a cleavable form at low protease concentrations, where $k_f \gg k_{obs}$ (Figure 1A). More generally, for all proteins (including non-two-state proteins with folding intermediates), monitoring proteolysis under conditions where any unfolded protein is immediately cleaved (i.e., cleavage is faster than refolding) reveals the rate of the first unfolding step and can be used to characterize cleavable, partially unfolded forms^{41,42} (i.e., at high protease concentrations, where $k_f \ll$

k_{obs} ; Figures 1A and S1). Gel electrophoresis provides a powerful readout of proteolysis, as resolving the size of cleavage products can also reveal structural information about unfolding intermediates^{41–43}; however, these assays have been limited to tens of variants at a time.

Here, we present a microfluidic assay that leverages native proteolysis to enable high-throughput measurements of protein unfolding rates (SPARKfold [simultaneous proteolysis assay revealing kinetics of folding]). SPARKfold monitors proteolytic cleavage of fluorescently labeled, surface-immobilized proteins over time under conditions in which unfolding is rate limiting, allowing the precise determination of 1,104 protein unfolding rate constants in less than 2 days. To demonstrate the power and potential of SPARKfold, we applied it to measure unfolding rates for

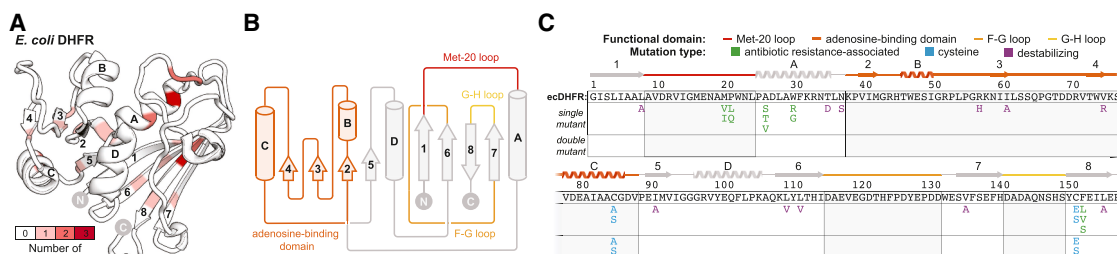


Figure 2. Overview of ecDHFR mutant library

(A) Cartoon schematic of the ecDHFR structure (PDB: 6CXK) showing the sites and numbers of mutations in this study (see C for mutations); α helices (letters) and β sheets (numbers) are labeled in ascending order from N- to C-terminal. Residues in the structure are colored by the number of substitutions at each position (white = 0, light pink = 1, dark pink = 2, and red = 3). (B) Cartoon schematic of the topology diagram for the ecDHFR structure showing locations of α helices (cylinders), β sheets (arrows), Met-20 loop (red), adenosine-binding domain (dark orange), F-G loop (light orange), and G-H loop (yellow). (C) Schematic showing each mutation in this study along a linear topology map indicating antibiotic-resistance-associated (green), cysteine (blue), or thermodynamically destabilizing (purple) substitutions; mutations and their associated references are also listed in Table S1.

a library of 31 dihydrofolate reductase (DHFR) orthologs and variants.⁴² SPARKfold measurements agreed with those from traditional native proteolysis using both gel electrophoresis and activity-based assays; across the entire library, they revealed that single DHFR substitutions can increase unfolding rates by up to 30-fold. Comparisons with prior measurements of the impact of mutations on thermodynamic stability^{42,44–47} allowed us to carry out phi analysis, which suggested that the central β sheet is disrupted in the cleavable intermediate state. SPARKfold can be applied across a wide variety of systems to quantify unfolding rates and characterize transition states, revealing biophysical principles governing protein-folding pathways, informing the development of engineered proteins with enhanced kinetic stability, and uncovering how mutations drive protein misfolding and aggregation in disease.

RESULTS

SPARKfold enables high-throughput expression, immobilization, purification, and native proteolysis of protein constructs

High-throughput functional characterization of large protein libraries requires the ability to recombinantly express and purify 100s–1,000s of protein variants in parallel, expose expressed proteins to protease, and follow cleavage of the expressed proteins over time. To accomplish this, SPARKfold employs microfluidic devices with 1,104 valved reaction chambers to express and immobilize fluorescently tagged protein variants, introduce protease, and monitor loss of fluorescence due to proteolysis over time (Figure 1C). Each device contains 24 parallel channels that contain 46 individual $\sim$ 1-nL reaction chambers (1,104 chambers total) (Figures 1C and S2); chambers are grouped into 6 individually addressable “blocks” of 4 channels each, making it possible to assay up to 6 different conditions (e.g., varying [urea], pH, salt, ligands, etc.) and perform experiments for all constructs and conditions across the device simultaneously (Figure 1C). Each chamber contains DNA and reaction compartments with three valves that control fluid flow: (1) neck valves that separate the DNA and reaction compartments, (2) sandwich valves that isolate reaction chambers from one another, and (3) button valves that enable selective surface

patterning and protein purification and provide precise temporal control over when the immobilized proteins are exposed to proteases.^{48,49}

To program each chamber with a unique protein variant, we align microfluidic devices to arrays of spotted plasmids that each encode a construct composed of a peptide or protein variant sandwiched between an N-terminal SNAP tag and a C-terminal monomeric EGFP tag,⁵⁰ connected via flexible glycine/serine linkers (Figures 1D and S3). After alignment, we actuate valves to pattern a circular surface patch within each chamber with BSA conjugated to benzyl-guanine (BG)⁵¹ and then introduce reagents for parallelized cell-free protein synthesis within each chamber (see STAR Methods). After expression, the protein constructs are irreversibly recruited to the patterned surfaces via a covalent bond formed between the cysteine in the active site of the N-terminal SNAP tag and BSA as the BG group is cleaved. Subsequently closing the valves protects the surface-immobilized constructs from flow, making it possible to wash and purify all expressed variants in parallel and to replace the expression buffers and reagents with the conditions desired for the unfolding experiments (Figure 1D). After quantifying the expressed variants, we introduce a protease to all chambers simultaneously and monitor the loss of EGFP fluorescence over time (Figure 1E). Here, we cleaved using thermolysin, a relatively non-specific protease that preferentially cleaves hydrophobic residues expected to be exposed during unfolding. Faster unfolding results in faster loss of the native protein population and EGFP fluorescence (Figure 1E). Tens of devices can be fabricated in hours, and a single experiment with one device can produce up to 1,104 protein unfolding curves over an additional day.

DHFR variant library provides a test application of SPARKfold

DHFR catalyzes the conversion of dihydrofolate to tetrahydrofolate (a key precursor in several metabolic pathways) and is a target for various small-molecule therapeutics used to treat cancer and bacterial infections. *E. coli* DHFR (ecDHFR) has long served as a model system for investigating protein-folding pathways^{42,46,52} and is ideal for our studies: (1) DHFR is a small (159-amino-acid), monomeric protein with no metals or disulfide bonds (Figures 2A and 2B), simplifying variant library synthesis

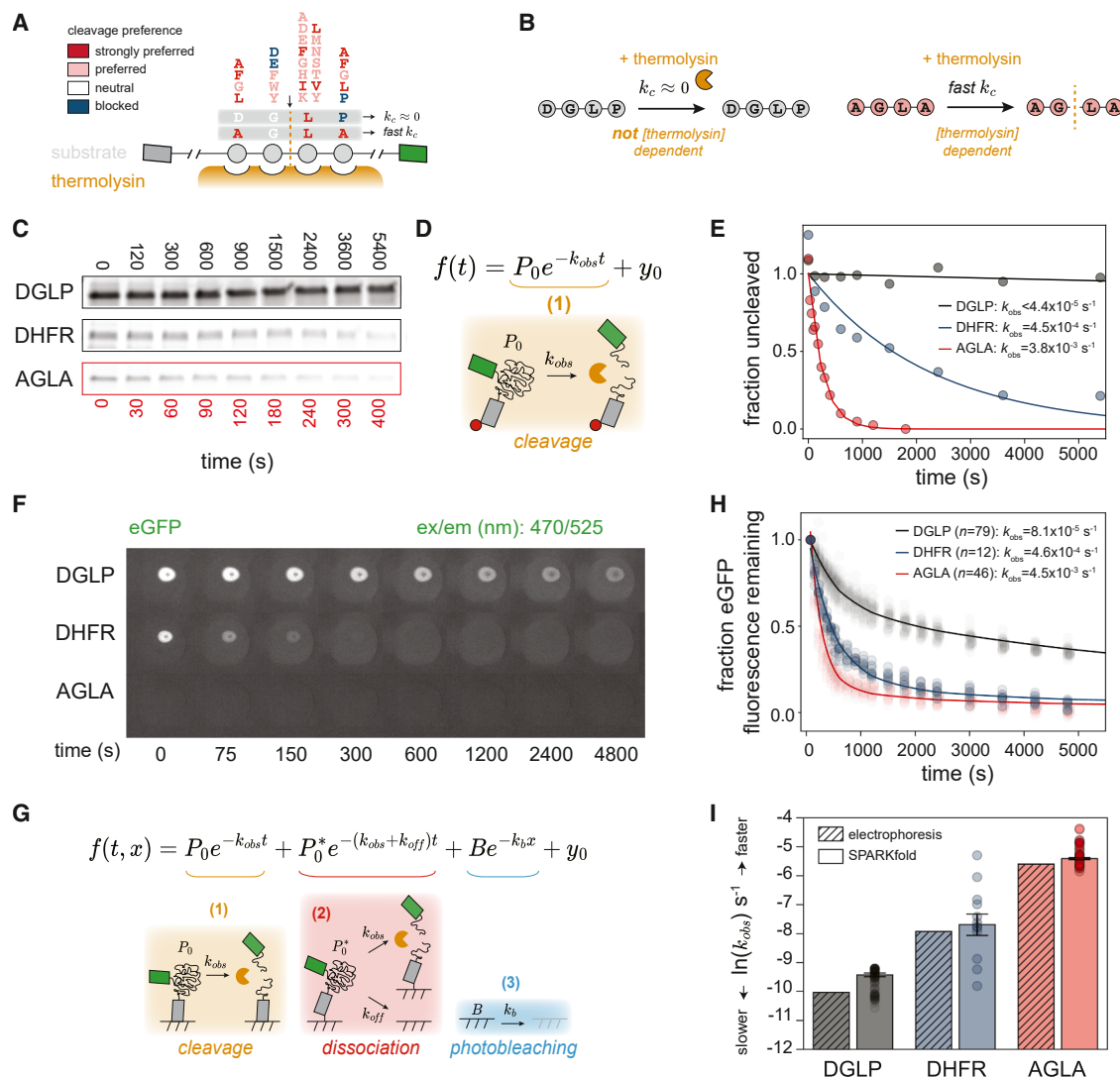


Figure 3. SPARKfold detects sequence-specific proteolytic cleavage and unfolding rates

(A) Schematic showing thermolysin (orange) specificity for residues flanking the substrate cleavage site (gray circles).⁷¹⁻⁷³ N-terminal SNAP (gray rectangle) and C-terminal EGFP (green rectangle) tags indicate cleavage-site orientation; substitutions are indicated as strongly preferred (dark red), preferred (pink), neutral (white), or blocked (blue) for cleavage.

(B) Negative (Asp-Gly-Leu-Pro [DGLP], left) and positive (Ala-Gly-Leu-Ala [AGLA], right) linear peptide controls for thermolysin cleavage.

(C) Montage of fluorescence images of SDS-PAGE gels showing the amount of full-length protein constructs remaining after native proteolysis at the same time points (from 0 to 10,800 s) for WT ecDHFR and control substrates AGLA and DGLP; the N-terminal SNAP tag is labeled with SNAP-Surface 649 (see [Figure S6](#) for full gels).

(D) Single-exponential cleavage model for the fraction of uncleaved protein (f) as a function of time (t) observed via electrophoresis; P_0 represents the initial protein population, k_{obs} is the observed rate of proteolysis, and y_0 is a constant offset.

(E) Fraction of uncleaved protein remaining as a function of time quantified from electrophoresis for DGLP, WT ecDHFR, and AGLA from one representative gel each (see [Figure S6](#) for all gels). Markers represent normalized intensities at a given time point, and lines indicate fits to single-exponential decay model; the measured cleavage rate for DGLP represents a lower limit.

(F) Example SPARKfold fluorescence images of individual chambers set at the same brightness settings showing EGFP intensities as a function of time after exposure to thermolysin protease; images are typical examples of 79, 12, and 46 chambers each for DGLP, WT ecDHFR, and AGLA.

(G) Composite model describing observed EGFP fluorescence (f) remaining as a function of time (t) and image acquisition number (x) that includes loss of fluorescence due to cleavage (orange), dissociation of a weakly bound protein population (red), and photobleaching (blue). P_0 represents the initial amount of strongly bound protein, k_{obs} represents the observed rate of proteolysis, P_0^* represents the initial amount of weakly bound protein, k_{off} represents the dissociation rate, B is the initial device and slide fluorescence, k_p is the photobleaching rate, and γ_0 is a constant offset.

(H) Fraction of EGFP fluorescence remaining as a function of time after SPARKfold native proteolysis for all chambers with DGLP, WT ecDHFR, and AGLA. Markers represent median normalized “button” intensities, and lines indicate fits to the plotted data using the cleavage, dissociation, and photobleaching model.

(legend continued on next page)

and expression; (2) there are many existing datasets describing DHFR stability and unfolding rate constants to compare with our observed data^{46,53}; (3) multiple high-resolution crystal structures (with and without bound ligands)^{54,55} facilitate interpretation of effects on residue interactions; and (4) multiple groups have determined that the energy landscape contains distinct kinetic intermediates that are selectively cleaved by proteases.^{56–63}

To develop and validate SPARKfold, we generated a library of 31 DHFR orthologs and variants: wild-type (WT) human DHFR (hDHFR), WT ecDHFR, and 29 ecDHFR variants, including 12 associated with antibiotic resistance,^{64–67} 11 previously shown to be thermodynamically destabilizing,^{42,47,68} and 6 with cysteine substitutions commonly used in biochemical studies and thought to have no impact on folding or activity⁴⁵ (Figures 2A–2C; Table S1). The ecDHFR mutations are found in α helices, β sheets, and unstructured loops throughout the enzyme, including in the adenosine-binding domain and Met-20 loop that regulate catalysis⁶⁹ (Figures 2B and 2C).

Off-chip proteolysis for 7 constructs provides benchmark measurements and establishes that unfolding is rate limiting

Accurately quantifying protein unfolding rate constants via SPARKfold requires (1) that protein unfolding is the rate-limiting step for cleavage and (2) that rate constants derived from measurements of protein cleavage on-chip match those from traditional gel-based measurements.^{42,70} To quantify thermolysin cleavage rates on- and off-chip, we designed positive- and negative-control constructs with N-terminal SNAP and C-terminal EGFP tags linked to linear peptides with flanking residues predicted to be either strongly preferred (Ala-Gly-Leu-Ala [AGLA]) or disfavored (Asp-Gly-Leu-Pro [DGLP]) thermolysin cleavage sites, based on prior work^{71–73} (Figures 3A and 3B). As these linear peptides do not fold, measured loss of fluorescence for these constructs should directly report on thermolysin cleavage only and quantify SPARKfold's maximum potential dynamic range (Figures 3A and 3B).

To provide benchmark off-chip measurements of unfolding rates, we used native proteolysis in solution coupled with gel electrophoresis to visualize cleavage time courses for the AGLA and DGLP positive- and negative-control constructs and the 5 DHFR orthologs and variants with a wide range of expected unfolding rates (hDHFR, ecDHFR, and the previously characterized ecDHFR variants L8A, W30R, and V75R)^{42,44–47,67,68,70,74} (Figures 3C and S4–S6; Table S2). As expected, fluorescent band intensities for SNAP domains labeled with SNAP-Surface 649 dyes corresponding to the full-length constructs decreased with increasing proteolysis time (Figures 3C and S6). Observed losses of intensity over time were well fit by a single-exponential decay, consistent with the expected simple Lumry-Eyring model reaction scheme⁷ (Figures 1D, 3D, 3E, and S7). DGLP cleavage was slower than could be reliably detected ($k_c < 4.4 \times 10^{-5} \text{ s}^{-1}$), AGLA was cleaved at a much faster rate con-

stant ($k_c = 3.8 \times 10^{-3} \text{ s}^{-1}$), and ecDHFR cleavage was within this range ($k_u = 4.5 \times 10^{-4} \text{ s}^{-1}$) (Figures 3C, 3D, and S7). To maximize assay dynamic range, these constructs (and all constructs used throughout the paper) included two mutations that removed thermolysin-preferred cleavage sites within the unstructured terminal regions of the SNAP and EGFP tags to reduce rates of background cleavage (EGFP V2G/SNAP L181E) (Figure S8). Measurements of cleavage time courses as a function of thermolysin concentration confirmed that protein unfolding was rate limiting: while cleavage rates for the positive-control linear peptide containing a thermolysin-preferred cleavage site (AGLA) increased linearly with thermolysin concentration ($k_c = 0.043(\text{g/L})^{-1} \cdot \text{s}^{-1}$ for thermolysin concentrations ranging from 0.012 to 0.5 g/L) (Figure S9), cleavage rates for an ecDHFR variant with previously measured fast unfolding rates (V75R) did not vary with increasing thermolysin (thus, unfolding is expected to be rate limiting across the library). Additional control experiments verified that thermolysin activity was not inhibited by components of the cell-free protein synthesis mixture or by the presence of expressed ecDHFR constructs (Figure S10). Plate-based activity measurements before and after native proteolysis (Figure S11) yielded a loss in activity consistent with gel electrophoresis estimates of the amount of protein cleaved ($R^2 = 0.84$, Figure S12; Table S2), confirming that proteolysis was degrading properly folded proteins (presumably by cleaving molecules that transitioned to the unfolded state over the course of the assay).

SPARKfold detects sequence-dependent proteolytic cleavage and returns accurate unfolding rate constants

To assess SPARKfold feasibility, we then tested whether SPARKfold could successfully express and detect sequence-specific proteolytic cleavage of surface-immobilized linear positive (AGLA) and negative (DGLP) control constructs and ecDHFR via loss of fluorescence. As a control, we first confirmed that thermolysin activity was not inhibited by conditions required for on-chip assays, including (1) high concentrations of BSA (above those used to prevent non-specific sticking within devices) and (2) overnight incubations in Tygon tubing at room temperature (Figure S13). Chambers containing plasmids encoding all 3 constructs showed substantial EGFP intensities while empty chambers did not, confirming successful expression and surface immobilization of SNAP- and EGFP-tagged proteins without cross-contamination (Figure 3F). As expected, fluorescence images revealed that EGFP fluorescence was lost fastest for the positive-control (AGLA) construct, consistent with sequence-specific proteolysis of surface-immobilized proteins (Figure 3F). However, while solution assays of DGLP revealed minimal cleavage, chambers containing this construct showed a considerable decrease in EGFP fluorescence over time (Figure 3F). This behavior was well fit by a functional form derived for a reaction scheme that includes photobleaching (proportional to the number of times the sample is exposed to light; see below) and two populations of surface-immobilized molecules: (1) a strongly bound population that

For each construct, the median k_{obs} value is reported with the sample size (n). Data are normalized across multiple chambers here to show trends; intensities and fits for each chamber are available as supplemental data.

(I) Comparison between natural-log-transformed k_{obs} ($\ln(k_{\text{obs}})$) values for DGLP (gray), WT ecDHFR (blue), and AGLA (red). Bars for traditional measurements (striped) indicate data from a single gel. Bars for SPARKfold (solid) measurements indicate median values across all chambers with the standard error (see also Table S2). Scatter points display values from individual chambers; the number of chambers for each measurement is as noted in (H).

is cleaved by thermolysin and (2) a weakly bound population that can either be cleaved or slowly dissociate from the surface (Figures 3G and 3H; see supplemental information for the derivation).

$$I(t, x) = P_0 e^{-k_{\text{obs}} t} + P_0^* e^{-(k_{\text{obs}} + k_{\text{off}}) t} + B e^{-k_b x} + y_0 \quad (\text{Equation 1})$$

Here, P_0 is the starting population of strongly bound molecules, k_{obs} is the observed cleavage rate constant, P_0^* is the starting population of weakly bound molecules, k_{off} represents their dissociation rate constant, B is the initial fluorescence of molecules that photobleach over the course of the experiment, k_b is the photobleaching rate constant, t is time, x is the number of images acquired at a given time point, and y_0 represents the initial constant offset. Each SPARKfold experiment includes empty and DGLP-containing chambers throughout, making it possible to quantify photobleaching and slow dissociation rate constants independently. We therefore determined observed cleavage rates for DHFR variants via a 3-step process. First, we determined a median amplitude and rate constant for photobleaching by fitting observed intensity vs. image number from all blank chambers to a single-exponential decay (Equation 2; Figure S14):

$$I_{\text{blank}}(x) = B e^{-k_b x} + y_0 \quad (\text{Equation 2})$$

Next, we determined a median rate constant for slow dissociation by fitting observed intensity vs. time from negative cleavage control (DGLP)-containing chambers with the photobleaching rate held constant and assuming $k_{\text{off}} \gg k_{\text{obs}}$ (as for a non-cleavable substrate; Equation 3; Figure S15):

$$I_{\text{DGLP}}(t, x) = P_0 e^{-k_{\text{obs}} t} + P_0^* e^{-k_{\text{off}} t} + B_{\text{blank}} e^{-k_b \text{blank} x} + y_0. \quad (\text{Equation 3})$$

Finally, we fit observed rate constants for cleavage (and thus inferred unfolding rate constants, assuming unfolding is rate limiting) for AGLA- and ecDHFR-containing chambers using Equation 1 and the median photobleaching and slow dissociation rate constants obtained above (Figures 3H, S16, and S17):

$$I(t, x) = P_0 e^{-k_{\text{obs}} t} + P_0^* e^{-(k_{\text{obs}} + k_{\text{off, DGLP}}) t} + B_{\text{blank}} e^{-k_b \text{blank} x} + y_0. \quad (\text{Equation 4})$$

With this procedure, median SPARKfold-returned cleavage rate constants across all chambers for DGLP, WT ecDHFR, and AGLA agreed with those determined by traditional gel-based native proteolysis within 2-fold over a 150-fold range of k_{obs} values (Figure 3I).

SPARKfold enables parallel unfolding rate measurements for the DHFR variant library

Next, we applied SPARKfold to quantify unfolding rates for all 31 orthologs and variants within the DHFR library, each at high redundancy to ensure excellent statistics. Chambers containing ecDHFR variants demonstrated higher expression and surface immobilization compared with empty chambers, although these levels were lower than those in chambers with control peptide constructs (Figure 4A). Based on these distributions, we set a minimum intensity threshold to exclude chambers with poor expression from downstream analysis (Figure 4B), yielding data for 90% of DHFR orthologs and variants (28/31) with 5 or

more chambers passing this threshold across experiments (Figure 4B). To estimate unfolding rate constants for each variant, we again (1) quantified rate constants for photobleaching from blank chambers (Figure S14), (2) used these photobleaching rate constants and data from DGLP-containing chambers to quantify surface dissociation rate constants (Figure S15), and then (3) fit observed fluorescence over time to a kinetic model that includes these processes and unfolding (Figures S16–S18). Raw curves for every chamber within each of 2 replicate experiments and compiled summary data across each experiment (Figure S19) are available as Supplementary Files and in an associated OSF repository.⁷⁵ Median cleavage rates (for control peptides) and unfolding rates (for DHFR variants) replicated reasonably well across experiments ($R^2 = 0.84$, Figure 4C; Table S3) and agreed with those determined with gel-based native proteolysis (Figure 4D, $R^2 = 0.95$). In general, constructs with relatively low initial EGFP intensities yielded rate-constant distributions with the highest variance (Figures 4E and S20), suggesting that future improvements in on-chip immobilization could improve assay precision (see discussion). Printing 31 variants with many (~10 to 50) replicates per variant allowed us to iteratively down-sample to probe how measurement error depended on the number of chambers sampled per variant; 5 chambers per variant returned root-mean-square error (RMSE) values for natural-log-transformed observed rates of $<0.5 \text{ s}^{-1}$ and coefficients of variation of $<5\%$ (Figures S21–S23).

Measured unfolding rate constants across ecDHFR variants varied by ~20-fold. Most variants (17/28) unfolded significantly faster than WT, and no variants unfolded more slowly (as determined by bootstrapped significance estimation with Bonferroni correction for multiple comparisons; Figures 4E and S24; Table S3). Proteolysis rates for the WT and 7 variants of ecDHFR were previously measured via traditional native proteolysis with a gel electrophoresis readout⁴²; however, these initial measurements used a lower concentration of thermolysin (to probe the equilibrium between the native and cleavable intermediate states rather than unfolding rate constants) and did not consider possible impacts of product inhibition. SPARKfold-returned measurements show similar sequence-dependent trends and agree with published rates when corrected for expected product inhibition⁷⁰ (Figure S25).

Substituting charged residues within the hydrophobic core drives large increases in unfolding rate

The suite of unfolding rates returned by SPARKfold provides an opportunity to search for sequence and structural features that shape the DHFR unfolding pathway. Projecting SPARKfold-measured unfolding rates onto the ecDHFR structure revealed no simple relationship between the spatial distribution of variants and their impact on kinetic stability (Figure 5A). For a mutation to alter the unfolding rate, it must alter the height of the barrier between the native state and the transition state. Because the structure of the transition-state ensemble is less well known, we consider impacts on k_u in the context of the interactions observed in the native state. Consistent with expected impacts of disrupting residue-packing interactions and secondary-structure formation, mutations to residues within β sheets, α helices, and unstructured loops increased median k_u by ~6-, 3-, and 2-fold, respectively (Figure 5B). Nevertheless, the magnitude of observed effects within secondary-structure elements varied

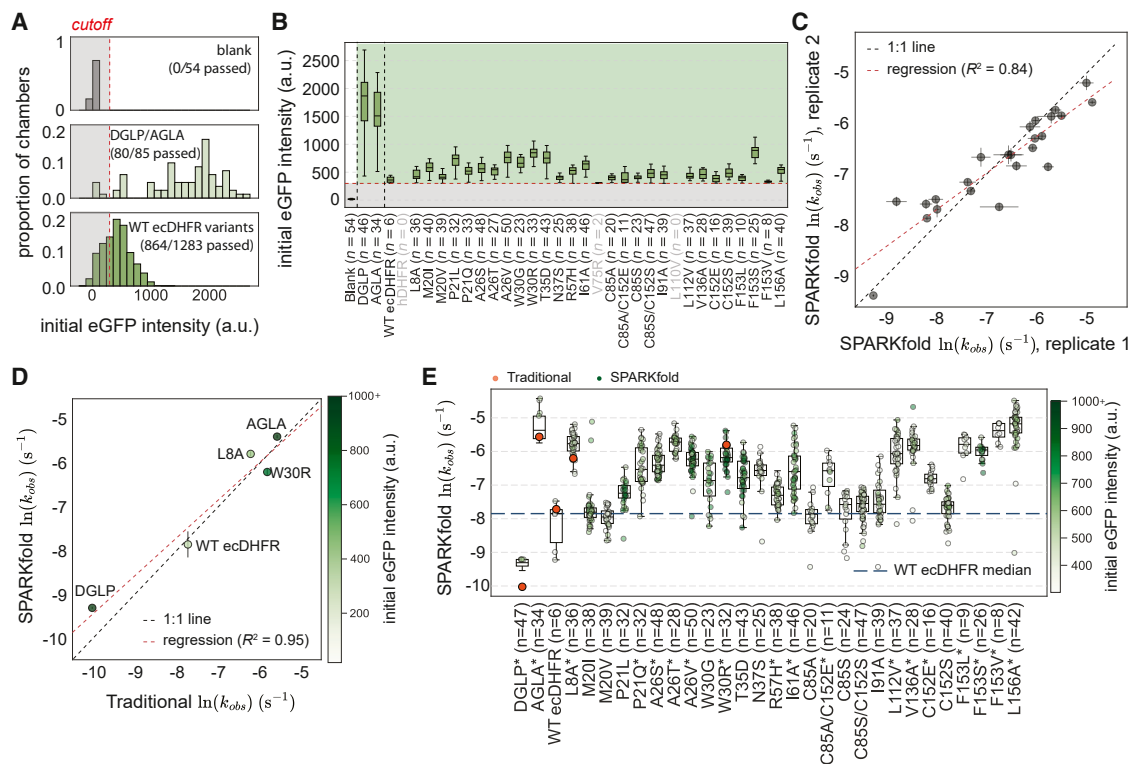


Figure 4. SPARKfold expression and unfolding rate measurement for library variants

(A) Histograms of per-chamber initial background-subtracted EGFP intensities for blank (top), control peptide (DGLP/AGLA) (middle), and DHFR mutant (bottom) chambers; annotation indicates the sample size (n) for each. The red dashed line represents a threshold of 156 a.u. to filter out chambers with low expression. Each histogram is normalized to show the proportion of data within each bin.

(B) Boxplot of initial background-subtracted EGFP chamber intensities by construct. Boxes span the interquartile range (IQR), lines inside boxes represent the median, and whiskers extend to $1.5 \times \text{IQR}$ beyond the box. The red dashed line marks the expression threshold (156 a.u.); shading distinguishes blank chambers (left), control peptide chambers (DGLP/AGLA) (middle), and protein chambers (DHFR mutants) (right). Construct labels include sample sizes (n).

(C) Scatterplot comparing natural-log-transformed k_{obs} ($\ln(k_{\text{obs}})$) values for two SPARKfold experimental replicates. Each marker represents a construct, error bars indicate the standard error of the median $\ln(k_{\text{obs}})$ for each replicate, the red dashed line indicates the linear fit to the plotted data ($R^2 = 0.87$), and the black dotted line indicates the 1:1 relationship.

(D) Scatterplot comparing $\ln(k_{\text{obs}})$ values for SPARKfold and traditional gel-based native proteolysis experiments across replicates. Each marker represents a construct, error bars indicate the per-replicate standard error of the median $\ln(k_{\text{obs}})$, the red dashed line indicates the linear fit to the plotted data ($R^2 = 0.87$), and the black dotted line indicates the 1:1 relationship. The gradient color bar reflects the median initial background-subtracted EGFP intensity on a linear scale with higher values capped at 1,000, illustrating that experimental variance is highest for variants with low expression.

(E) Boxplot with an overlaid strip-plot comparing $\ln(k_{\text{obs}})$ values for SPARKfold and traditional gel-based native proteolysis experiments by construct. Boxes span the IQR, lines inside boxes represent the median, whiskers extend to $1.5 \times \text{IQR}$ beyond this range are shown as outliers. Strip-plot markers (green) display individual replicate values. The gradient color bar reflects the median initial background-subtracted EGFP intensity on a linear scale, with higher values capped at 1,000. Markers (orange) show $\ln(k_{\text{obs}})$ values from traditional gel-based experiments, and the blue dashed line indicates the median $\ln(k_{\text{obs}})$ for WT ecDHFR. Construct labels include sample sizes (n) and Bonferroni-corrected significance markers ($p < 0.05$) compared with WT ecDHFR; values are also given in Table S3.

substantially depending on the tertiary contacts made within the native structure and the properties of the substituted residue (Figures 5A and 5B). Disrupting β sheet residues that directly contact other residues in the native structure generally increased k_u (e.g., L8, L112, V136, and L156, which pack against one another to form an antiparallel sheet and the hydrophobic core [Figures 5Aiii and 5B], and F153, which packs against A26 and forms pi-pi stacking interactions with W30 on an α helix [Figures 5Aiv and 5B]). At a given residue, the magnitude of the effect varied with the magnitude of the biochemical perturbation (e.g., substituting cysteine with a charged residue [C152E] was >2 -fold more deleterious than substituting it with a more conservative mutation [C152S]) (Figures 5Aiv and 5B). To test whether

measured unfolding rates for mutations in α helices could be explained by changes in helical propensity, we compared measured changes in k_u with previously determined changes in Gibbs free energy for helix formation⁷⁶ (see STAR Methods). For mutations within α helices, predicted changes in helical propensity accounted for $\sim 50\%$ of observed increases in unfolding rates (Figure 5C). The W30R mutation increases k_u more than predicted by the change in helical propensity, likely due to disrupted interactions with the β sheet residue F153 (Figures 5Aiv and 5C). Most residues within unstructured regions are found on the Met-20 loop. Although M20 mutations do not alter k_u , consistent with expectations for a flexible region, P21 substitutions on the same loop raise k_u by ~ 2 - to 4-fold, potentially by

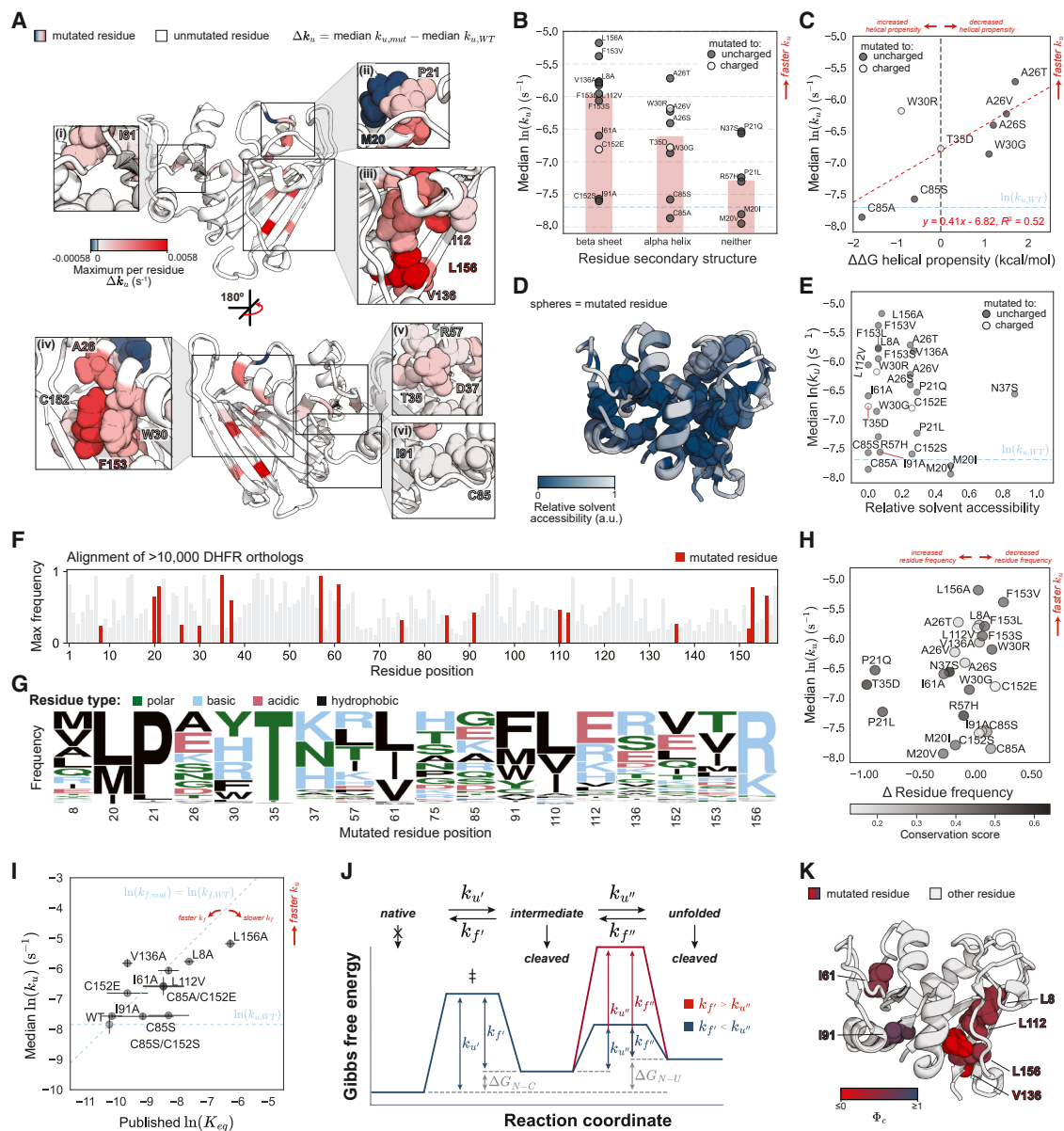


Figure 5. Analysis of ecDHFR unfolding rates based on structural and evolutionary features

(A) Entire and zoomed-in portions of the structure (PDB: 6CXK) showing the difference in measured unfolding rate between the mutant and WT ecDHFR (Δk_u) for the substitution with the largest effect at that position. Residue color ranges from no significant effect ($\Delta k_u < 0$; blue) to increased ($\Delta k_u > 0$; red) unfolding rates relative to WT, shown on a linear scale; at mutated positions, the native residue is displayed as spheres.

(B) Bar plot and overlaid strip-plot comparing natural-log-transformed k_u ($\ln(k_u)$) values for each construct based on the secondary structure at the substitution position. Strip-plot markers are colored based on mutation to an uncharged (filled circle) or charged (empty circle) residue. Bar heights indicate median $\ln(k_{\text{obs}})$ for substitutions made in β sheets (left), α helices (middle), and neither (right). The blue dashed line indicates the median $\ln(k_u)$ for WT ecDHFR.

(C) Scatterplot comparing median $\ln(k_u)$ values for ecDHFR variants in α helices with previously determined changes in Gibbs free energy for helix formation, $\Delta\Delta G$ for helical propensity (kcal/mol).⁷⁶ Markers are colored based on mutation to an uncharged (filled circle) or charged (empty circle) residue. The red dashed line indicates the linear fit to the plotted data ($R^2 = 0.54$), the blue dashed line indicates median $\ln(k_u)$ for WT ecDHFR, and the black dashed line indicates no change in helical propensity ($\Delta\Delta G = 0$ kcal/mol).

(D) Relative solvent accessibility of residues on the ecDHFR structure with residues mutated in this study shown as spheres. All ecDHFR residues are colored on a linear scale from low (dark blue) to high (light gray) relative solvent accessibility.

(E) Scatterplot comparing median $\ln(k_u)$ values for ecDHFR variants with relative solvent accessibility. Markers are colored based on mutation to an uncharged (filled circle) or charged (empty circle) residue. The blue dashed line indicates the median $\ln(k_u)$ for WT ecDHFR.

(F) Bar heights denote frequency of the most common residue at each position calculated from a multiple sequence alignment of >10,000 DHFR orthologs; red bars indicate mutated residue positions.

(G) Relative residue frequencies at each mutated residue position, colored based on residue type: polar (green), basic (blue), acidic (red), and hydrophobic (black).

(legend continued on next page)

enabling greater conformational flexibility (Figure 5B). Any insertion of a charged residue within the buried hydrophobic core increased unfolding rates by >2-fold (Figures 5D, 5E, and S26).

Evolutionary conservation does not correlate with observed impacts on unfolding rates

Enzymes must fold to function. Prior studies have reported correlations between mutational effects on stability and residue conservation within phylogenetic alignments and suggested the existence of a stability “threshold” below which fitness is compromised.^{77–84} However, it remains unclear whether such “thresholds” are dictated by thermodynamic stability, kinetic stability, both, or other factors. To investigate the relationship between evolutionary conservation and the impacts of mutations on folding rates, we aligned >10,000 DHFR orthologs and quantified residue frequencies at each position (Figures 5F and 5G). Impacts on k_u were largely uncorrelated with conservation, with many mutations to equally probable residues yielding substantial increases (Figure 5H). Although several previous studies have demonstrated that ecDHFR mutational landscapes vary substantially upon deletion of the quality-control protease Lon,^{85,86} we find no significant correlation when comparing either measured k_u or published K_{eq} values with Lon selection coefficients (Figure S27). These results underscore the complexity of biological selection. Although these results are consistent with prior DHFR studies that suggest that fitness depends on competition between chaperonins and proteases for access to unfolded intermediates,⁸⁵ establishing this relationship *in vitro* would require additional measurements of rate constants in the presence and absence of different proteases and chaperones.

Most DHFR variants reduce thermodynamic stability primarily by increasing the rate of unfolding

To gain insight into the DHFR unfolding pathway, we compared measured impacts of mutations on unfolding kinetics with previously published equilibrium stability measurements (Figure 5I; Table S4). For proteins that follow a two-state folding model, $K_{eq} = k_f/k_u$ such that mutations that cause the same change in K_{eq} and k_u (i.e., on the diagonal line, Figure 5I) can be inferred to have no impact on k_f . However, for proteins with intermediates, such as ribonuclease H (RNase H), thermodynamics and protease-cleavage-derived kinetics can be decoupled.⁸⁷ The DHFR folding pathway contains cleavable intermediate and unfolded states⁴⁶ such that measured proteolysis rates may either report on the kinetics of partial unfolding ($k_{u'}$) or global unfolding ($k_{u''}$), de-

pending on which step is rate limiting and which states are cleavable (Figure 5J); thus, thermodynamic stability and the measured proteolysis rate cannot be used to reliably infer the folding rate. Two mutations that pack against one another within the β sheet (V136A and L156A) increased apparent unfolding rate constants (Figures 5I and 5K); whereas V136A slightly decreased thermodynamic stability compared with WT, L156A decreased thermodynamic and kinetic stability similarly (Figures 5I and 5K).

Although the transition state cannot be isolated due to its transient nature, the bonds formed or broken in this state can be inferred via phi analysis. By comparing published K_{eq} and measured k_u values, we can determine phi (Φ_c), the ratio of the log-transformed mutation-induced change in the unfolding rate constant to the change in thermodynamic stability of the native form, thereby probing the structures of the transition-state intermediates along the unfolding pathway. Mutating residues that are unstructured within the unfolding transition (i.e., cleavable and high-energy) state alters the unfolding rate and the stability of the native form equally ($\Phi_c = 1$), whereas mutating residues that are structured in this transition state affects the stability of the native form alone but does not change the kinetic barrier ($\Phi_c = 0$; Figure 5K; see STAR Methods and Table S4). Projecting Φ_c values onto the structure suggests that the β sheet is primarily unfolded in the transition state while the adenosine-binding domain remains folded (Figure 5K), in agreement with previous conclusions based on the thermodynamic stability of the cleavable intermediate (ΔG_{N-C}).⁴²

DISCUSSION

Most published studies of protein stability focus on thermodynamic stability, likely because thermodynamic stability is more easily measured and predicted by existing assays and algorithms.¹¹ Yet in many cases, kinetic stability can be equally or more important in determining function. Secreted proteins that must function in harsh extracellular environments are often kinetically hyperstable, limiting their risk of degradation.⁵ Within cells, partial or complete unfolding can regulate signal transduction by modulating effective ligand-binding affinities and/or the accessibility of regions subject to posttranslational modifications.⁸⁸ Finally, proteins used in industrial or biotechnological applications face harsh solvent conditions and a high risk of irreversible denaturation.¹¹ Here, we present a technology (SPARKfold) that can be used to quantify kinetic stability under native (non-perturbative) conditions at scale.

(H) Scatterplot comparing median $\ln(k_u)$ values for ecDHFR variants with the change in frequency of the substituted residue relative to the native residue. Markers are colored based on conservation at the mutated residue position, ranging from less conserved (white) to more conserved (black) on a linear scale.

(I) Scatterplot comparing median $\ln(k_u)$ values for ecDHFR variants with $\ln(K_{eq})$, calculated from published free energy of unfolding (ΔG_u) values.^{43–47,68} Vertical error bars indicate the standard error of the median $\ln(k_{obs})$ for each replicate; horizontal error bars indicate the lower and upper limits of $\ln(K_{eq})$ calculated from the range of previously determined ΔG_u values; the blue dashed line indicates the median $\ln(k_u)$ for WT ecDHFR. The dashed diagonal line shows the expected correlation between k_u and K_{eq} if stability is only dependent on the unfolding rate; mutants to the left of the dashed line have a faster k_f , whereas mutants to the right have a slower k_f relative to WT ecDHFR.

(J) Gibbs free energy vs. reaction coordinate diagram for native, intermediate, and unfolded protein populations, where intermediate and unfolded proteins can be cleaved by a protease. The height of the energy barrier between the native and intermediate states depends on the unfolding rate ($k_{u'}$) and folding rate (k_f); the height of the energy barrier between the intermediate and unfolded states depends on the unfolding rate ($k_{u''}$) and folding rate (k_f). The change in Gibbs free energy (gray) is annotated between native and cleavable states (ΔG_{N-C}) and between native and unfolded states (ΔG_{N-U}). Example diagrams are shown under conditions where $k_f > k_{u'}$ (red) and $k_f < k_{u'}$ (blue).

(K) Calculated phi (Φ_c) values displayed on the ecDHFR structure (PDB: 6CXK). Mutated residues with thermodynamic stability data available to calculate Φ_c are displayed as spheres and colored from $\Phi_c \leq 0$ (red) to $\Phi_c \geq 1$ (blue) on a linear scale (see also Table S4).

In principle, other high-throughput peptide and protein production techniques (e.g., phage-, yeast-, cDNA-, and mRNA-display and array-based protein expression methods^{89–93}) could also be adapted to quantify unfolding rates at scale. Prior work by the Rocklin lab elegantly deployed proteolysis to measure thermodynamic folding stability for nearly 1 million small (40- to 72-amino-acid) *de novo*-designed protein domains.²⁵ To date, however, these experiments have not yet demonstrated that measurements are taking place under conditions in which the kinetic accessibility of the cleavable conformation is rate limiting, precluding the ability to accurately measure unfolding rates. Relative to many of these display-based techniques, SPARKfold offers 2 main advantages: (1) the ability to express larger protein constructs^{94,95} (as evidenced by the successful expression of the 593- and 776-amino-acid SNAP-ecDHFR-EGFP and PafA-EGFP proteins used here and in prior work) and (2) the ability to separately quantify additional biophysical properties for a given protein variant prior to proteolysis (e.g., measuring binding affinities and kinetics and/or catalytic activities via the previously published STAMMP [Simultaneous Transcription factor Affinity Measurements via Microfluidic Protein arrays],^{96,97} *k*-STAMMP [kinetic STAMMP],⁹⁸ STAMMPING [STAMMP-Protein Interaction Network Generator],⁹⁹ and HT-MEK [High-Throughput Microfluidic Enzyme Kinetics]^{100,101} techniques).

Previous measurements of equilibrium constants (K_{eq}) and unfolding rates (k_u) for ~30 single-domain, two-state proteins varied by 7 orders of magnitude.¹⁰² Here, most amino acid substitutions within ecDHFR increased unfolding rates by <2-fold, with the most deleterious substitutions increasing unfolding rates by ~20-fold. As mutations that dramatically alter kinetic stability would likely be subjected to strong negative selection, the small dynamic range observed here likely stems from our decision to profile variants observed in wild populations. Consistent with this model, several mutations with only minor impacts on unfolding rates were previously shown to affect global stability and ligand binding (e.g., trimethoprim-resistant and destabilized ecDHFR variants P21L and A26T^{67,103}).

DHFR tends to aggregate under the low-pH conditions required for hydrogen/deuterium exchange assays and to facilitate site-specific introduction of chemical modifications using thiol-reactive probes.¹⁰⁴ To prevent this aggregation, several prior studies have investigated cysteine-free double-mutant variants^{45,105–107} thought to retain similar activity, thermodynamic stability, and folding mechanisms to the WT protein.⁴⁵ Here, we find that the single- and double-cysteine mutations to alanine and serine have no statistically significant impact on unfolding rates, confirming the utility of these constructs for future studies of DHFR kinetic stability. The commonly used C152E mutation^{106,108,109} shows a small yet significant increase in unfolding rate (~3-fold, $p < 10^{-6}$; potentially due to a change in surface charge), suggesting that C152S should be used instead.

SPARKfold is compatible with any protein that (1) is not susceptible to proteolysis in the native state, (2) exposes a protease-compatible cleavage site when unfolding, and (3) refolds more slowly than the rate of proteolysis at feasible protease concentrations. For proteins that meet these criteria, SPARKfold identifies the rate at which the native folded population transitions to any folding intermediate with an accessible cleavage site, providing physiologically relevant information about the kinetics of inter-

change with aggregation- or cleavage-prone states. In this way, SPARKfold can nominate mutants for further detailed structural studies. Although we used thermolysin for all measurements here, additional measurements using proteases with differing specificities could provide additional information about local unfolding intermediates. If the globally unfolded state is the lowest-energy cleavable state, SPARKfold can provide quantitative information about global unfolding rates; otherwise, SPARKfold provides information about local unfolding. Here, the fastest and slowest unfolding rates that could be resolved were $k_{upper} = 0.013 \text{ s}^{-1}$ and $k_{lower} = 8.1 \times 10^{-5} \text{ s}^{-1}$. In future work, the upper limit of this dynamic range could be increased by altering the imaging setup to enable higher-throughput image acquisition. The lower limit of this dynamic range was imposed by observed slow dissociation of surface-immobilized constructs, similar to DGLP cleavage rates observed off-chip, suggesting cleavage of either the SNAP or EGFP tags, the flexible linkers, or the BSA adsorbed to the device surface. As variance in measured unfolding rates was inversely correlated with expression levels below a certain threshold, future optimization of variant DNA preparation, library printing, and on-chip expression conditions could boost expression and reduce this variability. We suspect that incomplete construct fragments containing the N-terminal SNAP tag and a fragment of DHFR observed on gels could occupy available binding sites, thereby reducing binding of the desired full-length protein and overall signal to noise.

Beyond measuring unfolding kinetics, SPARKfold lays the foundation for a wide variety of additional assays. SPARKfold enables precise measurement of small changes in k_u for many mutations at each residue position without disrupting the global structure. Thus, SPARKfold measurements can reveal mechanisms by which intermediate kinetics may impact a protein's cellular abundance or propensity to aggregate; measurements monitoring cleavage by multiple proteases with different cleavage preferences could further provide clues regarding intermediate structures. When combined with K_{eq} measurements, SPARKfold provides unique insights into folding intermediates suggested to be conserved across orthologs for complex enzymes.⁸⁸ As ligand binding stabilizes protein folded conformations, comparing unfolding rates measured in the presence and absence of a drug could provide a sensitive method for high-throughput quantification of binding of an unlabeled ligand. Finally, applying SPARKfold to varying linear (unfolded) sequences could provide a high-throughput method for systematically investigating the determinants of protease specificity.

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Polly Fordyce (pfordyce@stanford.edu).

Materials availability

Plasmids generated in this study have been deposited in AddGene.

Data and code availability

Raw data from all unfolding measurements have been deposited at the Open Science Framework: <https://doi.org/10.17605/OSF.IO/EGCU4>. Supplementary data files summarizing results per experiment and across experiments are available in the "Supplemental Files" folder within the same repository.

The original analytical code (Sparkfold) and input files used to generate the figures reported in this paper are available at the OSF (<https://doi.org/10.17605/OSF.IO/EGCU4>). Automation and processing software (AcqPack, ImageStitcher, ProcessingPack, and RunPack) were previously described^{100,110} and are available at the links provided in the [key resources table](#).

ACKNOWLEDGMENTS

This work was supported by NIH grant number R01GM064798 awarded jointly to D.H. and P.M.F.; P.M.F. is also a Chan Zuckerberg Biohub, San Francisco Investigator. B.A. was supported by an NSF Graduate Research Fellowship. The authors thank all members of the Fordyce and Herschlag labs for helpful feedback on the manuscript. In particular, we thank Professor Susan Marqusee and Shawn Costello for sharing their vast knowledge of protein folding and stability, Karl Krauth for his insights into mathematical modeling, Kara Brower for her help with microfluidic device design, and Matt DeJong for his conceptual help with protein surface immobilization.

AUTHOR CONTRIBUTIONS

Conceptualization, all authors; investigation, B.A. and F.S.; writing – original draft, all authors; writing – review and editing, all authors; and supervision and funding acquisition, P.M.F. and D.H.

DECLARATION OF INTERESTS

P.M.F. is a member of the *Cell Systems* advisory board and a co-founder of Velocity Bio.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

- [KEY RESOURCES TABLE](#)
- [METHODS DETAILS](#)
 - Cloning DHFR into PURExpress vector for cell-free expression
 - Optimization of the tags
 - Site-directed mutagenesis
 - Cell-free expression
 - Preparation of thermolysin
 - Test of loss of activity of thermolysin over time
 - Native proteolysis (gel-based assay)
 - Native proteolysis (activity-based assay)
 - Mold and device fabrication
 - Preparation of poly-L-lysine-coated slides
 - Plasmid printing and device alignment
 - Microscopy instrumentation
 - Conjugation of a SNAP substrate benzyl group (BG) to BSA
 - Surface patterning for construct immobilization
 - On-chip expression of double-tagged constructs
 - Surface immobilization of expressed constructs
 - Buffer exchange and thermolysin washing
 - Imaging and quantification of expressed constructs
 - Measuring unfolding rates on-chip
 - Testing statistical significance of measured effects
 - Estimation of time and cost for traditional vs. SPARKfold methods
 - ecDHFR variant analyses

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.cels.2026.101681>.

Received: January 6, 2025

Revised: May 18, 2026

Accepted: June 29, 2026

Published: July 29, 2026

REFERENCES

1. Englander, S.W., Mayne, L., and Krishna, M.M.G. (2007). Protein folding and misfolding: mechanism and principles. *Q. Rev. Biophys.* 40, 287–326. <https://doi.org/10.1017/S0033583508004654>.
2. Bai, Y., and Englander, S.W. (1996). Future directions in folding: The multi-state nature of protein structure. *Proteins* 24, 145–151. [https://doi.org/10.1002/\(SICI\)1097-0134\(199602\)24:2<145::AID-PROT1>3.0.CO;2-I](https://doi.org/10.1002/(SICI)1097-0134(199602)24:2<145::AID-PROT1>3.0.CO;2-I).
3. Ramirez-Alvarado, M. (2008). Principles of Protein Misfolding. In *Progress in Molecular Biology and Translational Science*, 84 (Elsevier), pp. 115–160. [https://doi.org/10.1016/S0079-6603\(08\)00404-2](https://doi.org/10.1016/S0079-6603(08)00404-2).
4. Fink, A.L. (1998). Protein aggregation: folding aggregates, inclusion bodies and amyloid. *Fold. Des.* 3, R9–R23. [https://doi.org/10.1016/S1359-0278\(98\)00002-9](https://doi.org/10.1016/S1359-0278(98)00002-9).
5. Plaza Del Pino, I.M., Ibarra-Molero, B., and Sanchez-Ruiz, J.M. (2000). Lower kinetic limit to protein thermal stability: A proposal regarding protein stability in vivo and its relation with misfolding diseases. *Proteins* 40, 58–70. [https://doi.org/10.1002/\(SICI\)1097-0134\(20000701\)40:1<58::AID-PROT80>3.0.CO;2-M](https://doi.org/10.1002/(SICI)1097-0134(20000701)40:1<58::AID-PROT80>3.0.CO;2-M).
6. Sanchez-Ruiz, J.M. (1992). Theoretical analysis of Lumry-Eyring models in differential scanning calorimetry. *Biophys. J.* 61, 921–935. [https://doi.org/10.1016/S0006-3495\(92\)81899-4](https://doi.org/10.1016/S0006-3495(92)81899-4).
7. Lumry, R., and Eyring, H. (1954). Conformation Changes of Proteins. *J. Phys. Chem.* 58, 110–120. <https://doi.org/10.1021/j150512a005>.
8. Colón, W., Church, J., Sen, J., Thibeault, J., Trasatti, H., and Xia, K. (2017). Biological Roles of Protein Kinetic Stability. *Biochemistry* 56, 6179–6186. <https://doi.org/10.1021/acs.biochem.7b00942>.
9. Jäger, M., Kelly, J.W., and Gruebele, M. (2024). Conservation of kinetic stability, but not the unfolding mechanism, between human transthyretin and a transthyretin-related enzyme. *Proc. Natl. Acad. Sci. USA* 121, e2315007121. <https://doi.org/10.1073/pnas.2315007121>.
10. Nixon, C.F., Lim, S.A., Sailer, Z.R., Zheludev, I.N., Gee, C.L., Kelch, B.A., Harms, M.J., and Marqusee, S. (2021). Exploring the Evolutionary History of Kinetic Stability in the α -Lytic Protease Family. *Biochemistry* 60, 170–181. <https://doi.org/10.1021/acs.biochem.0c00720>.
11. Sanchez-Ruiz, J.M. (2010). Protein kinetic stability. *Biophys. Chem.* 148, 1–15. <https://doi.org/10.1016/j.bpc.2010.02.004>.
12. Orosz, A., Wisniewski, J., and Wu, C. (1996). Regulation of *Drosophila* Heat Shock Factor Trimerization: Global Sequence Requirements and Independence of Nuclear Localization. *Mol. Cell. Biol.* 16, 7018–7030. <https://doi.org/10.1128/MCB.16.12.7018>.
13. Chan, D.C., Fass, D., Berger, J.M., and Kim, P.S. (1997). Core Structure of gp41 from the HIV Envelope Glycoprotein. *Cell* 89, 263–273. [https://doi.org/10.1016/S0092-8674\(00\)80205-6](https://doi.org/10.1016/S0092-8674(00)80205-6).
14. Bullough, P.A., Hughson, F.M., Skehel, J.J., and Wiley, D.C. (1994). Structure of influenza haemagglutinin at the pH of membrane fusion. *Nature* 371, 37–43. <https://doi.org/10.1038/371037a0>.
15. Carr, C.M., and Kim, P.S. (1993). A spring-loaded mechanism for the conformational change of influenza hemagglutinin. *Cell* 73, 823–832. [https://doi.org/10.1016/0092-8674\(93\)90260-W](https://doi.org/10.1016/0092-8674(93)90260-W).
16. Huber, R., and Carrell, R.W. (1989). Implications of the three-dimensional structure of alpha 1-antitrypsin for structure and function of serpins. *Biochemistry* 28, 8951–8966. <https://doi.org/10.1021/bi00449a001>.
17. Lee, C., Park, S.H., Lee, M.Y., and Yu, M.H. (2000). Regulation of protein function by native metastability. *Proc. Natl. Acad. Sci. USA* 97, 7727–7731. <https://doi.org/10.1073/pnas.97.14.7727>.
18. Hagai, T., and Levy, Y. (2010). Ubiquitin not only serves as a tag but also assists degradation by inducing protein unfolding. *Proc. Natl. Acad. Sci. USA* 107, 2001–2006. <https://doi.org/10.1073/pnas.0912335107>.
19. Carroll, E.C., Greene, E.R., Martin, A., and Marqusee, S. (2020). Site-specific ubiquitination affects protein energetics and proteasomal degradation. *Nat. Chem. Biol.* 16, 866–875. <https://doi.org/10.1038/s41589-020-0556-3>.
20. Swaney, D.L., Beltrao, P., Starita, L., Guo, A., Rush, J., Fields, S., Krogan, N.J., and Villén, J. (2013). Global analysis of phosphorylation and

- ubiquitylation cross-talk in protein degradation. *Nat. Methods* 10, 676–682. <https://doi.org/10.1038/nmeth.2519>.
21. Lee, J.M., Hammarén, H.M., Savitski, M.M., and Baek, S.H. (2023). Control of protein stability by post-translational modifications. *Nat. Commun.* 14, 201. <https://doi.org/10.1038/s41467-023-35795-8>.
22. Walker, E.J., Bettinger, J.Q., Welle, K.A., Hryhorenko, J.R., and Ghaemmighami, S. (2019). Global analysis of methionine oxidation provides a census of folding stabilities for the human proteome. *Proc. Natl. Acad. Sci. USA* 116, 6081–6090. <https://doi.org/10.1073/pnas.1819851116>.
23. Savitski, M.M., Reinhard, F.B.M., Franken, H., Werner, T., Savitski, M.F., Eberhard, D., Martinez Molina, D., Jafari, R., Dovega, R.B., Klaeger, S., et al. (2014). Tracking cancer drugs in living cells by thermal profiling of the proteome. *Science* 346, 1255784. <https://doi.org/10.1126/science.1255784>.
24. Rocklin, G.J., Chidyausiku, T.M., Goresnik, I., Ford, A., Houlston, S., Lemak, A., Carter, L., Ravichandran, R., Mulligan, V.K., Chevalier, A., et al. (2017). Global analysis of protein folding using massively parallel design, synthesis, and testing. *Science* 357, 168–175. <https://doi.org/10.1126/science.aan0693>.
25. Tsuboyama, K., Dauparas, J., Chen, J., Laine, E., Mohseni Behbahani, Y., Weinstein, J.J., Mangan, N.M., Ovchinnikov, S., and Rocklin, G.J. (2023). Mega-scale experimental analysis of protein folding stability in biology and design. *Nature* 620, 434–444. <https://doi.org/10.1038/s41586-023-06328-6>.
26. Atsavapranee, B., Stark, C.D., Sunden, F., Thompson, S., and Fordyce, P.M. (2021). Fundamentals to function: Quantitative and scalable approaches for measuring protein stability. *Cell Syst.* 12, 547–560. <https://doi.org/10.1016/j.cels.2021.05.009>.
27. Roder, H., and Shastry, M.R. (1999). Methods for exploring early events in protein folding. *Curr. Opin. Struct. Biol.* 9, 620–626. [https://doi.org/10.1016/S0959-440X\(99\)00015-9](https://doi.org/10.1016/S0959-440X(99)00015-9).
28. Kim, P.S., and Baldwin, R.L. (1982). Specific Intermediates in the Folding Reactions of Small Proteins and the Mechanism of Protein Folding. *Annu. Rev. Biochem.* 51, 459–489. <https://doi.org/10.1146/annurev.bi.51.070182.002331>.
29. Evans, P.A., and Radford, S.E. (1994). Probing the structure of folding intermediates. *Curr. Opin. Struct. Biol.* 4, 100–106. [https://doi.org/10.1016/S0959-440X\(94\)90066-3](https://doi.org/10.1016/S0959-440X(94)90066-3).
30. Matthews, C.R. (1993). PATHWAYS OF PROTEIN FOLDING. *Annu. Rev. Biochem.* 62, 653–683. <https://doi.org/10.1146/annurev.bi.62.070193.003253>.
31. Tanford, C., Aune, K.C., and Ikai, A. (1973). Kinetics of unfolding and refolding of proteins. 3. Results for lysozyme. *J. Mol. Biol.* 73, 185–197. [https://doi.org/10.1016/0022-2836\(73\)90322-7](https://doi.org/10.1016/0022-2836(73)90322-7).
32. Roder, H., and Colón, W. (1997). Kinetic role of early intermediates in protein folding. *Curr. Opin. Struct. Biol.* 7, 15–28. [https://doi.org/10.1016/S0959-440X\(97\)80004-8](https://doi.org/10.1016/S0959-440X(97)80004-8).
33. Deniz, A.A., Laurence, T.A., Beligere, G.S., Dahan, M., Martin, A.B., Chemla, D.S., Dawson, P.E., Schultz, P.G., and Weiss, S. (2000). Single-molecule protein folding: Diffusion fluorescence resonance energy transfer studies of the denaturation of chymotrypsin inhibitor 2. *Proc. Natl. Acad. Sci. USA* 97, 5179–5184. <https://doi.org/10.1073/pnas.090104997>.
34. Talaga, D.S., Lau, W.L., Roder, H., Tang, J., Jia, Y., DeGrado, W.F., and Hochstrasser, R.M. (2000). Dynamics and folding of single two-stranded coiled-coil peptides studied by fluorescent energy transfer confocal microscopy. *Proc. Natl. Acad. Sci. USA* 97, 13021–13026. <https://doi.org/10.1073/pnas.97.24.13021>.
35. Jia, Y., Talaga, D.S., Lau, W.L., Lu, H.S.M., DeGrado, W.F., and Hochstrasser, R.M. (1999). Folding dynamics of single GCN-4 peptides by fluorescence resonant energy transfer confocal microscopy. *Chem. Phys.* 247, 69–83. [https://doi.org/10.1016/S0301-0104\(99\)00127-5](https://doi.org/10.1016/S0301-0104(99)00127-5).
36. Ha, T., Enderle, T., Ogletree, D.F., Chemla, D.S., Selvin, P.R., and Weiss, S. (1996). Probing the interaction between two single molecules: fluorescence resonance energy transfer between a single donor and a single acceptor. *Proc. Natl. Acad. Sci. USA* 93, 6264–6268. <https://doi.org/10.1073/pnas.93.13.6264>.
37. Cieplak-Rotowska, M.K., Tamowski, K., Rubin, M., Fabian, M.R., Sonenberg, N., Dadlez, M., and Niedzwiecka, A. (2018). Structural Dynamics of the GW182 Silencing Domain Including its RNA Recognition motif (RRM) Revealed by Hydrogen-Deuterium Exchange Mass Spectrometry. *J. Am. Soc. Mass Spectrom.* 29, 158–173. <https://doi.org/10.1007/s13361-017-1830-9>.
38. Masson, G.R., Burke, J.E., Ahn, N.G., Anand, G.S., Borchers, C., Brier, S., Bou-Assaf, G.M., Engen, J.R., Englander, S.W., Faber, J., et al. (2019). Recommendations for performing, interpreting and reporting hydrogen deuterium exchange mass spectrometry (HDX-MS) experiments. *Nat. Methods* 16, 595–602. <https://doi.org/10.1038/s41592-019-0459-y>.
39. Chamberlain, A.K., Handel, T.M., and Marqusee, S. (1996). Detection of rare partially folded molecules in equilibrium with the native conformation of RNaseH. *Nat. Struct. Biol.* 3, 782–787. <https://doi.org/10.1038/nsb0996-782>.
40. Bai, Y., Sosnick, T.R., Mayne, L., and Englander, S.W. (1995). Protein Folding Intermediates: Native-State Hydrogen Exchange. *Science* 269, 192–197. <https://doi.org/10.1126/science.7618079>.
41. Park, C., and Marqusee, S. (2004). Probing the High Energy States in Proteins by Proteolysis. *J. Mol. Biol.* 343, 1467–1476. <https://doi.org/10.1016/j.jmb.2004.08.085>.
42. Kasper, J.R., Liu, P.F., and Park, C. (2014). Structure of a partially unfolded form of *Escherichia coli* dihydrofolate reductase provides insight into its folding pathway. *Protein Sci.* 23, 1728–1737. <https://doi.org/10.1002/pro.2555>.
43. Chang, Y., and Park, C. (2009). Mapping Transient Partial Unfolding by Protein Engineering and Native-State Proteolysis. *J. Mol. Biol.* 393, 543–556. <https://doi.org/10.1016/j.jmb.2009.08.006>.
44. Garvey, E.P., and Matthews, C.R. (1989). Effects of multiple replacements at a single position on the folding and stability of dihydrofolate reductase from *Escherichia coli*. *Biochemistry* 28, 2083–2093. <https://doi.org/10.1021/bi00431a018>.
45. Iwakura, M., Jones, B.E., Luo, J., and Matthews, C.R. (1995). A Strategy for Testing the Suitability of Cysteine Replacements in Dihydrofolate Reductase from *Escherichia coli*. *J. Biochem.* 117, 480–488. <https://doi.org/10.1093/oxfordjournals.jbchem.a124733>.
46. Touchette, N.A., Perry, K.M., and Matthews, C.R. (1986). Folding of dihydrofolate reductase from *Escherichia coli*. *Biochemistry* 25, 5445–5452. <https://doi.org/10.1021/bi00367a015>.
47. Samelson, A.J., Jensen, M.K., Soto, R.A., Cate, J.H.D., and Marqusee, S. (2016). Quantitative determination of ribosome nascent chain stability. *Proc. Natl. Acad. Sci. USA* 113, 13402–13407. <https://doi.org/10.1073/pnas.1610272113>.
48. Maerkl, S.J., and Quake, S.R. (2007). A Systems Approach to Measuring the Binding Energy Landscapes of Transcription Factors. *Science* 315, 233–237. <https://doi.org/10.1126/science.1131007>.
49. Fordyce, P.M., Gerber, D., Tran, D., Zheng, J., Li, H., DeRisi, J.L., and Quake, S.R. (2010). De novo identification and biophysical characterization of transcription-factor binding sites with microfluidic affinity analysis. *Nat. Biotechnol.* 28, 970–975. <https://doi.org/10.1038/nbt.1675>.
50. Zacharias, D.A., Violin, J.D., Newton, A.C., and Tsien, R.Y. (2002). Partitioning of Lipid-Modified Monomeric GFPs into Membrane Microdomains of Live Cells. *Science* 296, 913–916. <https://doi.org/10.1126/science.1068539>.
51. Meier, M., Sit, R., Pan, W., and Quake, S.R. (2012). High-Performance Binary Protein Interaction Screening in a Microfluidic Format. *Anal. Chem.* 84, 9572–9578. <https://doi.org/10.1021/ac302436y>.

52. Perry, K.M., Onuffer, J.J., Touchette, N.A., Herndon, C.S., Gittelman, M.S., Matthews, C.R., Chen, J.T., Mayer, R.J., Taira, K., and Benkovic, S.J. (1987). Effect of single amino acid replacements on the folding and stability of dihydrofolate reductase from *Escherichia coli*. *Biochemistry* 26, 2674–2682. <https://doi.org/10.1021/bi00384a004>.
53. Schnell, J.R., Dyson, H.J., and Wright, P.E. (2004). Structure, Dynamics, and Catalytic Function of Dihydrofolate Reductase. *Annu. Rev. Biophys. Biomol. Struct.* 33, 119–140. <https://doi.org/10.1146/annurev.biophys.33.110502.133613>.
54. Bolin, J.T., Filman, D.J., Matthews, D.A., Hamlin, R.C., and Kraut, J. (1982). Crystal structures of *Escherichia coli* and *Lactobacillus casei* dihydrofolate reductase refined at 1.7 Å resolution. I. General features and binding of methotrexate. *J. Biol. Chem.* 257, 13650–13662. [https://doi.org/10.1016/S0021-9258\(18\)33497-5](https://doi.org/10.1016/S0021-9258(18)33497-5).
55. Sawaya, M.R., and Kraut, J. (1997). Loop and Subdomain Movements in the Mechanism of *Escherichia coli* Dihydrofolate Reductase: Crystallographic Evidence. *Biochemistry* 36, 586–603. <https://doi.org/10.1021/bi962337c>.
56. Clark, A.C., and Frieden, C. (1999). The chaperonin GroEL binds to late-folding non-native conformations present in native *Escherichia coli* and murine dihydrofolate reductases. *J. Mol. Biol.* 285, 1777–1788. <https://doi.org/10.1006/jmbi.1998.2403>.
57. Arai, M., Kondrashkina, E., Kayatekin, C., Matthews, C.R., Iwakura, M., and Bilsel, O. (2007). Microsecond Hydrophobic Collapse in the Folding of *Escherichia coli* Dihydrofolate Reductase, an α/β -Type Protein. *J. Mol. Biol.* 368, 219–229. <https://doi.org/10.1016/j.jmb.2007.01.085>.
58. Arai, M., Iwakura, M., Matthews, C.R., and Bilsel, O. (2011). Microsecond Subdomain Folding in Dihydrofolate Reductase. *J. Mol. Biol.* 410, 329–342. <https://doi.org/10.1016/j.jmb.2011.04.057>.
59. Heidary, D.K., O'Neill, J.C., Roy, M., and Jennings, P.A. (2000). An essential intermediate in the folding of dihydrofolate reductase. *Proc. Natl. Acad. Sci. USA* 97, 5866–5870. <https://doi.org/10.1073/pnas.100547697>.
60. O'Neill, J.C., and Robert Matthews, C. (2000). Localized, stereochemically sensitive hydrophobic packing in an early folding intermediate of dihydrofolate reductase from *Escherichia coli*. *J. Mol. Biol.* 295, 737–744. <https://doi.org/10.1006/jmbi.1999.3403>.
61. Jones, S., and Thornton, J.M. (1995). Protein-protein interactions: A review of protein dimer structures. *Prog. Biophys. Mol. Biol.* 63, 31–65. [https://doi.org/10.1016/0079-6107\(94\)00008-W](https://doi.org/10.1016/0079-6107(94)00008-W).
62. Jennings, P.A., Finn, B.E., Jones, B.E., and Matthews, C.R. (1993). A re-examination of the folding mechanism of dihydrofolate reductase from *Escherichia coli*: Verification and refinement of a four-channel model. *Biochemistry* 32, 3783–3789. <https://doi.org/10.1021/bi00065a034>.
63. Frieden, C. (1990). Refolding of *Escherichia coli* dihydrofolate reductase: sequential formation of substrate binding sites. *Proc. Natl. Acad. Sci. USA* 87, 4413–4416. <https://doi.org/10.1073/pnas.87.12.4413>.
64. Watson, M., Liu, J.W., and Ollis, D. (2007). Directed evolution of trimethoprim resistance in *Escherichia coli*. *FEBS Journal* 274, 2661–2671. <https://doi.org/10.1111/j.1742-4658.2007.05801.x>.
65. Toprak, E., Veres, A., Michel, J.B., Chait, R., Hartl, D.L., and Kishony, R. (2011). Evolutionary paths to antibiotic resistance under dynamically sustained drug selection. *Nat. Genet.* 44, 101–105. <https://doi.org/10.1038/ng.1034>.
66. Cammarata, M., Thyer, R., Lombardo, M., Anderson, A., Wright, D., Ellington, A., and Brodbelt, J.S. (2017). Characterization of trimethoprim resistant *E. coli* dihydrofolate reductase mutants by mass spectrometry and inhibition by propargyl-linked antifolates. *Chem. Sci.* 8, 4062–4072. <https://doi.org/10.1039/C6SC05235E>.
67. Tamer, Y.T., Gaszek, I.K., Abdizadeh, H., Batur, T.A., Reynolds, K.A., Atilgan, A.R., Atilgan, C., and Toprak, E. (2019). High-Order Epistasis in Catalytic Power of Dihydrofolate Reductase Gives Rise to a Rugged Fitness Landscape in the Presence of Trimethoprim Selection. *Mol. Biol. Evol.* 36, 1533–1550. <https://doi.org/10.1093/molbev/msz086>.
68. Leontiev, V.V., Uversky, V.N., and Gudkov, A.T. (1993). Comparative stability of dihydrofolate reductase mutants *in vitro* and *in vivo*. *Protein Eng.* 6, 81–84. <https://doi.org/10.1093/protein/6.1.81>.
69. Osborne, M.J., Venkitakrishnan, R.P., Dyson, H.J., and Wright, P.E. (2003). Diagnostic chemical shift markers for loop conformation and substrate and cofactor binding in dihydrofolate reductase complexes. *Protein Sci.* 12, 2230–2238. <https://doi.org/10.1110/ps.03219603>.
70. Kasper, J.R., Andrews, E.C., and Park, C. (2014). Product inhibition in native-state proteolysis. *PLoS One* 9, e111416. <https://doi.org/10.1371/journal.pone.0111416>.
71. Ligné, T., Pauthe, E., Monti, J.P., Gacel, G., and Larreta-Garde, V. (1997). Additional data about thermolysin specificity in buffer- and glycerol-containing media. *Biochim. Biophys. Acta* 1337, 143–148. [https://doi.org/10.1016/S0167-4838\(96\)00142-2](https://doi.org/10.1016/S0167-4838(96)00142-2).
72. Keil, B. (1992). Specificity of Proteolysis (Springer). <https://doi.org/10.1007/978-3-642-48380-6>.
73. Heinrikson, R.L. (1977). [20] Applications of thermolysin in protein structural analysis. In *Methods in Enzymology*, 47 (Academic Press), pp. 175–189. [https://doi.org/10.1016/0076-6879\(77\)47022-8](https://doi.org/10.1016/0076-6879(77)47022-8).
74. Kampmeyer, C., Larsen-Ledet, S., Wagnkilde, M.R., Michelsen, M., Iversen, H.K.M., Nielsen, S.V., Lindemose, S., Caregnato, A., Ravid, T., Stein, A., et al. (2022). Disease-linked mutations cause exposure of a protein quality control decon. *Structure* 30, 1245–1253.e5. <https://doi.org/10.1016/j.str.2022.05.016>.
75. Atsavapranee, B., and SPAR. (2024). Kfold_DHFR. <https://doi.org/10.17605/OSF.IO/NAWME>.
76. O'Neil, K.T., and DeGrado, W.F. (1990). A Thermodynamic Scale for the Helix-Forming Tendencies of the Commonly Occurring Amino Acids. *Science* 250, 646–651. <https://doi.org/10.1126/science.2237415>.
77. Lehmann, M., Pasamontes, L., Lassen, S.F., and Wyss, M. (2000). The consensus concept for thermostability engineering of proteins. *Biochim. Biophys. Acta* 1543, 408–415. [https://doi.org/10.1016/S0167-4838\(00\)00238-7](https://doi.org/10.1016/S0167-4838(00)00238-7).
78. Sánchez, I.E., Tejero, J., Gómez-Moreno, C., Medina, M., and Serrano, L. (2006). Point Mutations in Protein Globular Domains: Contributions from Function, Stability and Misfolding. *J. Mol. Biol.* 363, 422–432. <https://doi.org/10.1016/j.jmb.2006.08.020>.
79. Campos, L.A., Garcia-Mira, M.M., Godoy-Ruiz, R., Sanchez-Ruiz, J.M., and Sancho, J. (2004). Do Proteins Always Benefit from a Stability Increase? Relevant and Residual Stabilisation in a Three-state Protein by Charge Optimisation. *J. Mol. Biol.* 344, 223–237. <https://doi.org/10.1016/j.jmb.2004.09.047>.
80. Pielak, G.J., Auld, D.S., Beasley, J.R., Betz, S.F., Cohen, D.S., Doyle, D.F., Finger, S.A., Fredericks, Z.L., Hilgen-Willis, S., and Saunders, A.J. (1995). Protein thermal denaturation, side-chain models, and evolution: amino acid substitutions at a conserved helix-helix interface. *Biochemistry* 34, 3268–3276. <https://doi.org/10.1021/bi00010a017>.
81. Steipe, B., Schiller, B., Plückthun, A., and Steinbacher, S. (1994). Sequence Statistics Reliably Predict Stabilizing Mutations in a Protein Domain. *J. Mol. Biol.* 240, 188–192. <https://doi.org/10.1006/jmbi.1994.1434>.
82. Godoy-Ruiz, R., Perez-Jimenez, R., Ibarra-Molero, B., and Sanchez-Ruiz, J.M. (2005). A Stability Pattern of Protein Hydrophobic Mutations that Reflects Evolutionary Structural Optimization. *Biophys. J.* 89, 3320–3331. <https://doi.org/10.1529/biophysj.105.067025>.
83. Godoy-Ruiz, R., Perez-Jimenez, R., Ibarra-Molero, B., and Sanchez-Ruiz, J.M. (2004). Relation Between Protein Stability, Evolution and Structure, as Probed by Carboxylic Acid Mutations. *J. Mol. Biol.* 336, 313–318. <https://doi.org/10.1016/j.jmb.2003.12.048>.
84. Godoy-Ruiz, R., Ariza, F., Rodriguez-Larrea, D., Perez-Jimenez, R., Ibarra-Molero, B., and Sanchez-Ruiz, J.M. (2006). Natural Selection for Kinetic Stability Is a Likely Origin of Correlations between Mutational Effects on Protein Energetics and Frequencies of Amino Acid

- Occurrences in Sequence Alignments. *J. Mol. Biol.* 362, 966–978. <https://doi.org/10.1016/j.jmb.2006.07.065>.
85. Bershtein, S., Mu, W., Serohijos, A.W.R., Zhou, J., and Shakhnovich, E.I. (2013). Protein Quality Control Acts on Folding Intermediates to Shape the Effects of Mutations on Organismal Fitness. *Mol. Cell* 49, 133–144. <https://doi.org/10.1016/j.molcel.2012.11.004>.
86. Thompson, S., Zhang, Y., Ingle, C., Reynolds, K.A., and Kortemme, T. (2020). Altered expression of a quality control protease in *E. coli* reshapes the in vivo mutational landscape of a model enzyme. *eLife* 9, e53476. <https://doi.org/10.7554/eLife.53476>.
87. Lim, S.A., Hart, K.M., Harms, M.J., and Marqusee, S. (2016). Evolutionary trend toward kinetic stability in the folding trajectory of RNases H. *Proc. Natl. Acad. Sci. USA* 113, 13045–13050. <https://doi.org/10.1073/pnas.1611781113>.
88. Mitrea, D.M., Yoon, M.-K., Ou, L., and Kriwacki, R.W. (2012). Disorder-function relationships for the cell cycle regulatory proteins p21 and p27. *Biol. Chem.* 393, 259–274. <https://doi.org/10.1515/hsz-2011-0254>.
89. Roberts, R.W., and Szostak, J.W. (1997). RNA-peptide fusions for the in vitro selection of peptides and proteins. *Proc. Natl. Acad. Sci. USA* 94, 12297–12302. <https://doi.org/10.1073/pnas.94.23.12297>.
90. Hanes, J., and Plückthun, A. (1997). In vitro selection and evolution of functional proteins by using ribosome display. *Proc. Natl. Acad. Sci. USA* 94, 4937–4942. <https://doi.org/10.1073/pnas.94.10.4937>.
91. Layton, C.J., McMahon, P.L., and Greenleaf, W.J. (2019). Large-Scale, Quantitative Protein Assays on a High-Throughput DNA Sequencing Chip. *Mol. Cell* 73, 1075–1082.e4. <https://doi.org/10.1016/j.molcel.2019.02.019>.
92. Ratnikov, B., Cieplak, P., and Smith, J.W. (2009). High Throughput Substrate Phage Display for Protease Profiling. *Methods Mol. Biol.* 539, 93–114. https://doi.org/10.1007/978-1-60327-003-8_6.
93. Jung, S., Honegger, A., and Plückthun, A. (1999). Selection for improved protein stability by phage display. *J. Mol. Biol.* 294, 163–180. <https://doi.org/10.1006/jmbi.1999.3196>.
94. Wang, H., and Liu, R. (2011). Advantages of mRNA display selections over other selection techniques for investigation of protein–protein interactions. *Expert Rev. Proteom.* 8, 335–346. <https://doi.org/10.1586/epr.11.15>.
95. Takahashi, T.T., Austin, R.J., and Roberts, R.W. (2003). mRNA display: ligand discovery, interaction analysis and beyond. *Trends Biochem. Sci.* 28, 159–165. [https://doi.org/10.1016/S0968-0004\(03\)00036-7](https://doi.org/10.1016/S0968-0004(03)00036-7).
96. Aditham, A.K., Markin, C.J., Mokhtari, D.A., DelRosso, N., and Fordyce, P.M. (2021). High-Throughput Affinity Measurements of Transcription Factor and DNA Mutations Reveal Affinity and Specificity Determinants. *Cell Syst.* 12, 112–127.e11. <https://doi.org/10.1016/j.cels.2020.11.012>.
97. Horton, C.A., Alexandari, A.M., Hayes, M.G.B., Marklund, E., Schaepe, J.M., Aditham, A.K., Shah, N., Suzuki, P.H., Shrikumar, A., Afek, A., et al. (2023). Short tandem repeats bind transcription factors to tune eukaryotic gene expression. *Science* 381, eadd1250. <https://doi.org/10.1126/science.add1250>.
98. Hastings, R., Aditham, A.K., DelRosso, N., Suzuki, P.H., and Fordyce, P.M. (2025). Mutations to transcription factor MAX allosterically increase DNA selectivity by altering folding and binding pathways. *Nat. Commun.* 16, 636. <https://doi.org/10.1038/s41467-024-55672-2>.
99. DelRosso, N., Suzuki, P.H., Griffith, D., Lotthammer, J.M., Novak, B., Kocalar, S., Sheth, M.U., Holehouse, A.S., Bintu, L., and Fordyce, P. (2024). High-throughput affinity measurements of direct interactions between activation domains and co-activators. Preprint at bioRxiv. <https://doi.org/10.1101/2024.08.19.608698>.
100. Markin, C.J., Mokhtari, D.A., Sunden, F., Appel, M.J., Akiva, E., Longwell, S.A., Sabatti, C., Herschlag, D., and Fordyce, P.M. (2021). Revealing enzyme functional architecture via high-throughput microfluidic enzyme kinetics. *Science* 373, eabf8761. <https://doi.org/10.1126/science.abf8761>.
101. Markin, C.J., Mokhtari, D.A., Du, S., Doukov, T., Sunden, F., Cook, J.A., Fordyce, P.M., and Herschlag, D. (2023). Decoupling of catalysis and transition state analog binding from mutations throughout a phosphatase revealed by high-throughput enzymology. *Proc. Natl. Acad. Sci. USA* 120, e2219074120. <https://doi.org/10.1073/pnas.2219074120>.
102. Maxwell, K.L., Wildes, D., Zarrine-Afsar, A., De Los Rios, M.A., Brown, A.G., Friel, C.T., Hedberg, L., Horng, J.C., Bona, D., Miller, E.J., et al. (2005). Protein folding: Defining a “standard” set of experimental conditions and a preliminary kinetic data set of two-state proteins. *Protein Sci.* 14, 602–616. <https://doi.org/10.1110/ps.041205405>.
103. Rodrigues, J.V., Bershtein, S., Li, A., Lozovsky, E.R., Hartl, D.L., and Shakhnovich, E.I. (2016). Biophysical principles predict fitness landscapes of drug resistance. *Proc. Natl. Acad. Sci. USA* 113, E1470–E1478. <https://doi.org/10.1073/pnas.1601441113>.
104. Jones, B.E., and Matthews, C.R. (1995). Early intermediates in the folding of dihydrofolate reductase from *Escherichia coli* detected by hydrogen exchange and NMR. *Protein Sci.* 4, 167–177. <https://doi.org/10.1002/pro.5560040204>.
105. Jing, C., and Cornish, V.W. (2013). A fluorogenic TMP-tag for high signal-to-background intracellular live cell imaging. *ACS Chem. Biol.* 8, 1704–1712. <https://doi.org/10.1021/cb300657r>.
106. Pucciarelli, S., Spina, M., Montecchia, F., Lupidi, G., Eleuteri, A.M., Fioretti, E., and Angeletti, M. (2005). Peroxynitrite-mediated oxidation of the C85S/C152E mutant of dihydrofolate reductase from *Escherichia coli*: functional and structural effects. *Arch. Biochem. Biophys.* 434, 221–231. <https://doi.org/10.1016/j.abb.2004.11.008>.
107. Moree, B., Connell, K., Mortensen, R.B., Liu, C.T., Benkovic, S.J., and Salafsky, J. (2015). Protein Conformational Changes Are Detected and Resolved Site Specifically by Second-Harmonic Generation. *Biophys. J.* 109, 806–815. <https://doi.org/10.1016/j.bpj.2015.07.016>.
108. Amici, M., Sagratini, D., Pettinari, A., Pucciarelli, S., Angeletti, M., and Eleuteri, A.M. (2004). 20S proteasome mediated degradation of DHFR: implications in neurodegenerative disorders. *Arch. Biochem. Biophys.* 422, 168–174. <https://doi.org/10.1016/j.abb.2003.12.014>.
109. Uedaira, H., Kidokoro, S., Iwakura, M., Honda, S., and Ohashi, S. (1990). Thermal transition of a mutated dihydrofolate reductase. *Thermochim. Acta* 163, 123–128. [https://doi.org/10.1016/0040-6031\(90\)80387-E](https://doi.org/10.1016/0040-6031(90)80387-E).
110. Longwell, S.A., and Fordyce, P.M. (2020). micrIO: an open-source auto-sampler and fraction collector for automated microfluidic input-output. *Lab Chip* 20, 93–106. <https://doi.org/10.1039/C9LC00512A>.
111. Park, C., and Marqusee, S. (2005). Pulse proteolysis: A simple method for quantitative determination of protein stability and ligand binding. *Nat. Methods* 2, 207–212. <https://doi.org/10.1038/nmeth740>.
112. Le, D.D., Shimko, T.C., Aditham, A.K., Keys, A.M., Longwell, S.A., Orenstein, Y., and Fordyce, P.M. (2018). Comprehensive, high-resolution binding energy landscapes reveal context dependencies of transcription factor binding. *Proc. Natl. Acad. Sci. USA* 115, E3702–E3711. <https://doi.org/10.1073/pnas.1715888115>.
113. Fordyce, P.M., Pincus, D., Kimmig, P., Nelson, C.S., El-Samad, H., Walter, P., and DeRisi, J.L. (2012). Basic leucine zipper transcription factor Hac1 binds DNA in two distinct modes as revealed by microfluidic analyses. *Proc. Natl. Acad. Sci. USA* 109, E3084–E3093. <https://doi.org/10.1073/pnas.1212457109>.
114. Brower, K., Puccinelli, R., Markin, C.J., Shimko, T.C., Longwell, S.A., Cruz, B., Gomez-Sjoberg, R., and Fordyce, P.M. (2018). An open-source, programmable pneumatic setup for operation and automated control of single- and multi-layer microfluidic devices. *HardwareX* 3, 117–134. <https://doi.org/10.1016/j.ohx.2017.10.001>.
115. Crooks, G.E., Hon, G., Chandonia, J.-M., and Brenner, S.E. (2004). WebLogo: a sequence logo generator. *Genome Res.* 14, 1188–1190. <https://doi.org/10.1101/gr.849004>.

STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Bacterial and Virus Strains		
NEB 5-alpha Competent <i>E. coli</i>	New England Biolabs	C29871
Chemicals, Peptides and Recombinant Proteins		
UltraPure BSA	ThermoFisher	AM2618
Phosphate Buffered Saline (10X)	Corning	46-013-CM
Trehalose Dihydrate	Sigma Life Science	T9531
Sodium Chloride	Sigma	204439
NHS-GLA-BG	New England Biolabs	S9151S
Thermolysin	Sigma	T7902
Borate buffer	LabChem	LC117001
EDTA	Ambion	AM9260G
SNAP-Surface 649	New England Biolabs	S9159S
Abz-AGLA-Nba	MedChemExpress	HY-P4406
Poly-L-lysine	Electron Microscopy Sciences	19320-A
Dihydrofolic acid	Sigma Aldrich	D7006
NADPH	Thermo Fischer	AC328742500
Calcium Chloride	Sigma	C2536
Critical Commercial Assays		
PURExpress <i>In Vitro</i> Protein Synthesis Kit	New England Biolabs	E6800L
RNAasin Ribonuclease Inhibitor	Promega	N2515
DpnI	New England Biolabs	R0176L
QIA Prep spin miniprep kit	QiaGen	27104
Deposited Data		
Raw unfolding measurements	This paper	OSF: https://doi.org/10.17605/OSF.IO/NAWME
Software and Code		
AcqPack	Longwell and Fordyce ¹⁰⁰	https://pypi.org/project/acqpack/
ImageStitcher	Markin et al. ¹⁰⁰	Github: https://github.com/FordyceLab/
ProcessingPack	Markin et al. ¹⁰⁰	Github: https://github.com/FordyceLab/ProcessingPack
RunPack	Markin et al. ¹⁰⁰	Github: https://github.com/FordyceLab/RunPack
Analytical code SPARKfold	This paper	OSF: 10.17605/OSF.IO/QRH8A

METHODS DETAILS

Cloning DHFR into PURExpress vector for cell-free expression

We cloned the gene coding for *E. coli* DHFR (ecDHFR), a thermolysin-preferred substrate (Ala-Gly-Leu-Ala; AGLA), and a thermolysin-resistant substrate (Asp-Gly-Leu-Pro; DGLP) into the PURExpress plasmid, which contains a C-terminal EGFP tag and an N-terminal SNAP tag, both attached with a flexible glycine/serine linker (GGGSGGG). We confirmed the sequence of the construct with Sanger sequencing, and all subsequent mutagenesis used the ecDHFR construct sequence as a plasmid template.

Optimization of the tags

To remove alternative expression start sites and stabilize the construct against proteolysis outside of the protein of interest, we used PeptideCutter to identify thermolysin cleavage sites in the unstructured regions between the protein and both the EGFP and SNAP tags. We performed site-directed mutagenesis (as described below) to generate the following mutations for all constructs: SNAP

L181E and EGFP M1G/V2G. For the DHFR construct, we introduced an additional mutation to remove the start codon (DHFR M1G). After confirming these optimized construct sequences with Sanger sequencing,⁷⁵ we used the two control constructs for thermolysin (SNAP-AGLA-EGFP and SNAP-DGLP-EGFP) to test the proteolysis resistance of the construct with and without stabilizing mutations, following rates of native proteolysis (described above).

Site-directed mutagenesis

We performed site-directed mutagenesis using the standard protocol for Pfu Turbo. We used the PCR product to transform chemically competent DH5a cells, which we made in-house following standard protocols, and plated them on LB agar carbenicillin plates. We picked single colonies and grew them in 10 mL LB with 50 µg/mL carbenicillin overnight on a rotor at 37°C. After pelleting the cells, we purified the plasmids using the GeneJet Plasmid Miniprep Kit (Thermo Scientific) and verified the sequence of the genes with Sanger sequencing.

Cell-free expression

We used the following protocol for the expression of double-tagged DHFR with the PURExpress *in vitro* synthesis kit (NEB): (1) we incubated 30 µL of Solution A with 22.5 µL of Solution B on ice for 45 minutes, (2) we added 1.5 µL of RNasin, 0.75 µL of 100 mM DTT, 19 µL of UltraPure water, and 1 µL of ~200 ng/µL plasmid, and then (3) we ran the *in vitro* transcription-translation reaction in a PCR machine for 2 hours at 30°C followed by 90 minutes at 23°C. After expression, we added 75 µL of 100 mM Tris-HCl (pH 8.0), 150 mM NaCl for unlabeled SNAP, and to label SNAP we added 10 nM benzyl guanine (BG)-conjugated Dynomics dye DY-649P1 (SNAP-Surface 649, NEB) and incubated at room temperature for 1 hour. We measured the amount of expressed protein based on EGFP fluorescence using a Denovix and calculated the concentration using an EGFP standard curve. For off-chip experiments, we buffer-exchanged the PURExpress in 0.5 mL Amicon filters (Sigma) 3 times with 100 mM Tris (pH = 8), 150 mM NaCl, and 1 mM DTT.

Preparation of thermolysin

We prepared 10 g/L thermolysin from *Geobacillus stearothermophilus* (Sigma) as described by Marqusee and Park.¹¹¹ We dissolved 100 mg of thermolysin in 10 mL of 2.5 M NaCl and 100 mM CaCl₂, separated the solution into 50 200-µL aliquots, and stored at -20°C.

Test of loss of activity of thermolysin over time

To assess if the observed rate of proteolysis was influenced by a change in the concentration of active thermolysin over time, we used a commercial protease substrate abz-AGLA-nBa (MedChemExpress) to measure the activity of thermolysin after incubation on the SPARKfold device, with BSA, or with PURExpress components. We dissolved the substrate in 30% acetic acid and stored it at -20°C. We incubated thermolysin (prepared as described above) in Tygon tubing (Saint-Gobain, AAD04103), BSA over a 10⁵-fold range (0.15 nM–15 µM), or with WT or V75R ecDHFR in PURExpress overnight. We then observed turnover kinetics in 384 plates at 314/417 nm on a Tecan M200, using 100 mM Tris-HCl (pH 8.0), 150 mM NaCl and 0.3–0.8 µM substrate.

Native proteolysis (gel-based assay)

To determine if the observed rate of proteolysis was limited by the K_{eq} between folded and unfolded protein or by the unfolding rate, we quantified the rate of DHFR proteolysis across a 20-fold range of thermolysin concentrations (0.06–0.5 g/L). We buffer-exchanged DHFR expressed with PURExpress into 100 mM Tris-HCl (pH 8.0) and 150 mM NaCl in 3K Amicon filters, diluting the buffer 100-fold. To the buffer-exchanged DHFR, we added different concentrations of thermolysin and collected timepoints by quenching 4 µL aliquots with 4 µL of 50 mM EDTA. We stored the aliquots on ice under tin foil until all time points had been collected. To each sample, we added 3 µL of Laemmli loading dye, ran the samples on denaturing gels (150V, 45 minutes), and imaged gels on a Typhoon FLA 9500 (for GFP: excitation blue laser (473 nm) and emission (>510 nm); for SNAP649: excitation red laser (635 nm) and emission (>665 nm)) or BioRad Versadoc. We used ImageJ to quantify the intensity of the gel bands corresponding to the full-length construct, plotted the concentration of full-length construct as a function of proteolysis time, and fit the data to a single exponential function. To ensure that observations that unfolding is rate-limiting for DHFR were not due to solubility issues or similar effects that would result in non-first order thermolysin cleavage kinetics, we also performed the above experiment with thermolysin substrate AGLA, varying the thermolysin concentration over a 100-fold range (0.012–1.0 g/L).

Native proteolysis (activity-based assay)

To provide an orthogonal readout of thermolysin cleavage rates (and verify that thermolysin was cleaving active protein that had unfolded), we quantified enzymatic activity after native proteolysis at multiple timepoints. First, we exposed DHFR constructs to thermolysin and collected timepoints as described for the denaturing gel assay above. We then added 4 µL of each DHFR sample to 200 µL of 100 mM Tris-HCl (pH 8.0), 150 mM NaCl, 1 mM DTT, 50 µM dihydrofolic acid (Sigma Aldrich D7006-10MG) and 60 µM NADPH (Thermo Fischer AC328742500) in a 96-well plate. Using a Tecan M200, we measured the DHFR activity in each sample by monitoring the change in substrate concentration at 340 nm, corresponding to conversion of substrate NADPH to product NADP⁺ by the active enzyme. For each sample, we fit observed fluorescence vs. proteolysis time to a single exponential decay model.

Mold and device fabrication

Flow and control molding masters were fabricated as previously described.^{112,113} Using these molds, we cast two-layer microfluidic devices using polydimethylsiloxane (PDMS) polymer (RS Hughes, RTV615). To generate control layers (Figure S1, orange), we combined 60 grams of PDMS components (1:5 ratio of cross-linker to base), mixed with a THINKY centrifugal mixer for 3 minutes at 2000 rpm, poured the mixture onto the molds, degassed in a vacuum chamber for 45 minutes, and then baked in a convection oven for 50 minutes at 80°C. After cutting and removing device control layers from the wafer, we added control inlets using a hole punch (SYNEO, CR0350255N20R4) mounted on a drill press (Technical Innovations). To generate flow layers (Figure S1, blue), we combined PDMS components (1:20 ratio of cross-linker to base), mixed with a THINKY centrifugal mixer for 3 minutes at 2000 rpm, and then spin-cast the PDMS onto the molds at 266 rpm for 10 seconds then 1750–1850 rpm for 75 seconds. After allowing flow layers to rest on a flat surface for 10 minutes at room temperature, we baked them for 40 minutes at 80°C. Using a stereoscope, we then aligned the control layers to flow layers on master molds. The assemblies were baked for an additional 40 minutes at 80°C and then cut from the molds with a scalpel. Flow inlets were created with the same catheter hole punch.

Preparation of poly-L-lysine-coated slides

PDMS devices do not adhere well to untreated glass. To functionalize glass surfaces to promote device adhesion and rounding of printed DNA droplets, we coated glass surfaces with poly-L-lysine. First, we washed 75 x 25 mm glass slides (Corning 2947) with Milli-Q H₂O followed by drying. Next, we diluted Poly-L-lysine 0.1% (Electron Microscopy Sciences, 19320-A) 10-fold into Milli-Q H₂O, added this solution to the glass slides, and allowed to shake for 1 hour at room temperature. Finally, we rinsed off the poly-L-lysine solution with Milli-Q H₂O, baked them in a convection oven for 1 hour at 60°C, and then stored them in the dark at room temperature until use.

Plasmid printing and device alignment

To prepare plasmid solutions for spotting onto slides, we mixed 4 μ L of each plasmid with 36 μ L of filtered print buffer solution (0.5 g/L BSA, 100 mM NaCl, 12 g/L Trehalose) in a 384 well plate (Thermo Scientific, AB-1055); we stored these solutions at -20°C when not in use. Prior to printing, we defrosted the plate, centrifuged at 4°C (2000g for 10 minutes), and then carefully removed the foil to prevent cross-contamination. We printed plasmids onto the poly-L-lysine-coated glass slides using a sciFLEXARRAYER S3 fitted with a PDC70 nozzle with Type coating (Sciencion). After printing, we aligned PDMS devices onto the printed arrays with each plasmid spot encapsulated by a different DNA compartment. Prior to use, we bonded the glass slides and PDMS devices together by baking for 3 hours at 95°C on a hotplate (Torrey Pines Scientific).

Microscopy instrumentation

We imaged all microfluidic devices using a Nikon Ti-S Microscope equipped with a motorized XY stage (Applied Scientific Instrumentation, MS-2000 XYZ stage), CMOS camera (Oxford Instruments, Andor Zyla 4.2 CMOS), and solid-state light source (Lumencor, Sola SE Light Engine), and automated filter turret with an EGFP filter set (Chroma Technology Corp., part no. 49002). We performed imaging with a 4X objective (CFI Plan Apochromat I 4X NA 0.20, Nikon) at 2x2 binning (1024x1024 pixels) with 500 ms exposure times. Throughout assays, we measured apparatus temperatures using a thermistor sensor (Thorlabs Inc., catalog no. TSP01, data logger; catalog no. TSP-TH NTC, external thermistor).

Conjugation of a SNAP substrate benzyl group (BG) to BSA

To covalently couple SNAP-substrate (benzyl guanine, BG) to BSA via N-hydroxysuccinimide (NHS) coupling, we dissolved NHS-GLA-BG (2 mg, S9151S, NEB) in 100 μ L DMF, slowly added 20 μ L of this solution to 1000 μ L of 5 g/L buffer-exchanged BSA in 10 mM sodium borate (pH 8.0), and stirred wrapped in tin foil overnight. The BG-BSA was divided into 50 μ L aliquots, flash frozen in liquid nitrogen, and stored at -80°C.

Surface patterning for construct immobilization

We connected Tygon tubing to a syringe at one end using a 23-gauge Luer connector (McMaster-Carr, 75165A684) and attached to a blunt steel pin at the other end (0.013 in ID x 0.025 in OD x 0.5 in length, New England Small Tube Corporation, NE-1310-02). With the reagents loaded into the syringe, we inserted the pin into the corresponding flow inlet on the microfluidic device. We then moved the other end of the tubing from the syringe to a custom pneumatic manifold, where fluid flow rates are controlled by pressure.¹¹⁴ To actuate control valves, we used a pneumatic control manifold¹⁰⁰ and Python software package (Github: <https://github.com/FordyceLab/RunPack>).¹⁰⁰

To prevent premature solubilization of the DNA spots by osmosis, we pressurized control lines for the button, sandwich, and neck valves (Figure S1) with 550mM NaCl in Milli-Q H₂O and remaining control lines with Milli-Q H₂O. We used pressures from 25–27 psi to control device pneumatic valves and 2.5–3 psi to introduce reagents.

After dead-end filling all control lines, we pressurized (closed) button and neck valves to protect a circular region on device surfaces from fluid and block flow to the DNA chambers. To reduce nonspecific binding, we then coated all remaining surfaces with ultrapure bovine serum albumin (BSA) (5 g/L, ThermoFisher Scientific, AM2616). After flowing the BSA through the inlet tree to waste for ~5 minutes, we opened the inlet valve to passivate all channels for ~5 minutes. We closed the outlet valve and continued to apply pressure for an additional 10 minutes to expel all air from the device flow layer, and flowed BSA over the chip for an additional 15 min.

Next, we flushed the device with phosphate buffered saline (10X stock, Corning, 46-013-CM; diluted to 1X in Milli-Q H₂O) for 15 minutes. We then opened button valves and introduced BG-BSA (prepared as described above) for 20 minutes to specifically pattern the region protected by this valve with BG-BSA, followed by another PBS wash for 10 minutes.

On-chip expression of double-tagged constructs

We solubilized printed DNA spots encoding double-tagged constructs and expressed protein using the PURExpress expression system (New England Biolabs Inc., catalog no. E6800L). To prepare the expression mixture, we mixed 60 μ L of PURExpress Solution A with 45 μ L of Solution B and allowed this mixture to incubate on ice for 30–45 minutes. Immediately before flowing the reaction mixture into the chip, we added 3 μ L of 40 U/L RNasin ribonuclease inhibitor (Promega Corporation, catalog no. N2515), 3 μ L of 100 mM DTT, and 39 μ L of nuclease-free water, and mixed gently. With the neck valves closed, we flowed this expression mixture over the device for 15 minutes to ensure that PURExpress solution fully filled all channels. To introduce the PURExpress mixture into all expression chambers and solubilize the DNA, we closed the chip outlet valve, opened the neck valves, and increased the flow pressure from to 4.0–4.5 psi, dead-end filling the DNA chambers under pressure. We monitored this process with a stereo microscope until chambers were 90–95% full and then closed the button, neck, and sandwich valves, isolating adjacent protein chambers from one another and preventing premature binding of constructs in the case of any leakage between expression chambers within the flow path. We then placed the slide bearing the device onto a pre-warmed aluminum hot plate (Torrey Pines Scientific) and allowed constructs to express for 2 hours at 30°C. Following expression, we removed devices from heat and incubated them for 90 minutes at room temperature in the dark to allow for maturation of the EGFP fluorophore.

Surface immobilization of expressed constructs

After construct expression, we removed the device/slide assembly from the hot plate and mounted it on the automated Nikon Ti-S microscope (components described above). To bind expressed constructs to the BG-BSA-patterned regions of the glass, we closed sandwich valves (to isolate adjacent reaction chambers from one another), opened button valves (to expose binding surfaces), and then opened the neck valve, allowing diffusion of expressed constructs into the reaction chambers. We allowed proteins to form covalent bonds between the SNAP tag and the BG-patterned surface for 30–60 min. During binding, we imaged chambers across the device in the EGFP channel to monitor binding, then closed button valves after observing minimal change in construct concentration.

Buffer exchange and thermolysin washing

Accurately measuring unfolding rates requires that chambers be free of contaminating constructs from other chambers and cell-free expression components, which can nonspecifically bind to device or slide surfaces. After closing button valves to protect immobilized protein, we opened sandwich valves to enable flow throughout the device, washed DNA and reaction compartments with Tris buffer (100 mM Tris-HCl (pH 8.0), 150 mM NaCl, 1 mM DTT) for 1 hour, and then closed the neck valves to isolate reaction compartments from DNA compartments. To dislodge any nonspecifically bound proteins from underneath the button valves, we opened and closed the button valves 10 times, flowed Tris buffer for 10 minutes, then repeated this process 2 more times. With the button and neck valves closed, we increased the control pressure to 28–30 psi to protect the construct bound underneath the button from premature exposure to thermolysin. We then flowed 0.1 g/L thermolysin (prepared as described above) onto the device for 10 minutes to remove any cleavable material outside of the button regions.

Imaging and quantification of expressed constructs

To quantify construct expression and proteolytic degradation, we imaged microfluidic devices using an EGFP filter set (Chroma Technology Corp., part no. 49002) with acquisition times of 500 ms per image. Using a Python software package (<https://pypi.org/project/acqpack/>),¹ each device was imaged using 2x2 binning and typically required 49 tiled images with 10% overlap. To assign mutant identities to each chamber, we stitched the overlapped tiles, divided each stitched image into 24x46 sub-images (stamps) centered on and containing a single chamber (using the 4 corner chambers as reference positions), determined the identity of the mutant at each position using a reference list, and then grouped images by mutant to process timecourses. Prior to the start of each proteolysis assay, we quantified starting EGFP intensities by: (1) identifying the centroid positions of button valve regions within each stamp using a combination of a sparse grid search (to identify the circular region of maximal summed fluorescence intensity) and a dense local grid search (to refine this position), (2) quantifying the median intensity from all pixels within a fixed radius of 15 pixels (larger than the physical radius of the button), and then (3) subtracting the local background by calculating the median intensity of pixels in an annulus concentric with, abutting, and larger than the button bounding circle. For fitting photobleaching rates, intensities were not background-subtracted.

Measuring unfolding rates on-chip

To quantify rates of proteolytic degradation, we: (1) loaded 0.1 g/L thermolysin into each chamber, (2) closed sandwich valves, (3) opened the buttons (to expose surface-immobilized protein to thermolysin), and (4) started automated image acquisition with a Python software package (<https://pypi.org/project/acqpack/>).¹¹⁰ Images were collected from time point 0 to 10,000 s; images were acquired more frequently at the beginning of the assay (to capture fast cleavage for constructs such as AGLA) and less frequently at the end of the assay (to capture slow cleavage for constructs such as DGLP while limiting photobleaching). As low-expressing mutants yielded unfolding rates with higher variance, data were filtered to only report values for variants with >5 chambers with measured

initial intensities > 300. All Jupyter notebooks and input text files required to perform all analyses and create all figures in the manuscript are available in an OSF repository (OSF: osf.io/egcu4/).

Testing statistical significance of measured effects

To assess the statistical significance of measured differences in unfolding rates, we performed bootstrapped hypothesis testing for each pairwise comparison between the set of measured unfolding rates for a given mutant vs. the WT construct. The null hypothesis for each test was that there was no difference in unfolding rates. To account for multiple comparisons, we applied a Bonferroni correction to adjust for the significance threshold by calculating a Bonferroni-corrected p -value threshold (α/N , where $\alpha = 0.05$ and N was the total number of mutants tested). Mutants with significant p -values were classified as having a statistically different unfolding rate compared to the WT.

Estimation of time and cost for traditional vs. SPARKfold methods

In a traditional assay, transforming the DNA encoding the desired protein mutants into bacteria and plating the cells to allow colony growth requires about 1.5 days, followed by 2 days to grow the cell cultures. Lysing cells and purifying the target proteins require 1 additional day. Buffer exchanging to transfer the protein into a compatible buffer for downstream assays takes 0.5 days. Unfolding assays, which can involve approaches with plate reader and SDS-PAGE gel readouts, require about 2 days, totaling 7 days. Assuming that 5 mutants can be prepared and assayed in parallel, 500 mutants can be studied in 700 days. If the salary for one person is \$62,500 per year (reflecting typical rates for graduate students and postdoctoral researchers), and there are 250 workdays per year, the labor cost for two people to complete 700 days of work each is \$350,000. In contrast, SPARKfold can simultaneously express and purify 500 mutants on a microfluidic chip within just 1 day, followed by buffer washes and unfolding assays completed in the following day. Given the same salaries and workdays, the cost for 2 people to perform this 2-day assay is \$1,000, a savings of >100 fold.

ecDHFR variant analyses

For ecDHFR mutations made in alpha helices, we determined the degree to which unfolding rates could be explained by predicted changes in helical propensity by plotting measured unfolding rates vs. predicted $\Delta\Delta G$ helical propensity values (calculated as ΔG for substituted residue - ΔG for native residue, using previously-reported amino acid helical propensities⁷⁶) and performed a linear regression. The relative solvent accessibility (RSA) of residues in the ecDHFR protein was calculated using PyMOL's `get_sasa_relative` function. This function computes the relative solvent accessible surface area for each residue, with values ranging from 0.0 (fully buried) to 1.0 (fully exposed). To assess ecDHFR conservation, we created a multiple sequence alignment (MSA) from PFAM (PFAM: PF00186), used BioPython to align each sequence to the reference species and remove gaps, and then calculated a frequency matrix of amino acid occurrences across the positions. We visualized conservation by plotting the maximum frequency at each ecDHFR position and creating sequence logos using WebLogo.¹¹⁵ For the Φ analysis, we calculated the free energy for partial unfolding ($\Delta\Delta G_{C-N}^\circ$) using median SPARKfold values for variant and WT ecDHFR, then determined Φ by using this measured $\Delta\Delta G_{C-N}^\circ$ and previously published measurements of free energy for global unfolding ($\Delta\Delta G_{U-N}^\circ$),^{42,44–47,68} as described by Park.⁴²